

Group & Ring Theory MATP33: Course Summary of Important Results

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Lecture 0 — Background Results

Theorem 1 (First Isomorphism Theorem). *Let $f : G \rightarrow H$ be a group homomorphism. Then $\ker f \trianglelefteq G$ and $\operatorname{im} f \cong G/\ker f$.*

Theorem 2 (Second Isomorphism Theorem). *Let G be a group, $H \leq G$, and $N \trianglelefteq G$. Then $HN \leq G$, $H \cap N \trianglelefteq H$, and $HN/N \cong H/(H \cap N)$.*

Theorem 3 (Third Isomorphism Theorem). *If $K \trianglelefteq G$ and $K \leq L \trianglelefteq G$, then $(G/K)/(L/K) \cong G/L$.*

Theorem 4 (Correspondence Theorem). *Let $N \trianglelefteq G$. Every subgroup of G/N is of the form H/N where $N \leq H \leq G$, and conversely if $N \leq H \leq G$ then $H/N \leq G/N$.*

Theorem 5 (Lagrange). *If $H \leq G$ and G is finite, then $|H|$ divides $|G|$.*

Theorem 6 (Fundamental Theorem of Finitely Generated Abelian Groups). *Let A be a finitely generated abelian group. Then*

$$A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z},$$

where $s \geq 0$ and $a_1 \mid a_2 \mid \cdots \mid a_r$ are non-zero non-units in $\mathbb{Z}$. This decomposition is unique.

Lecture 1

Theorem 7 (1.10). *Let G be a group and X a set. Then:*

- (i) *If X is a G -set, then the action of G on X induces a homomorphism $\varphi : G \rightarrow S_X$;*
- (ii) *Any homomorphism $\varphi : G \rightarrow S_X$ induces an action of G on X .*

Proof. (i) Define $\varphi : G \rightarrow S_X$ by $(\varphi(a))(x) = a \cdot x$ for $a \in G$ and $x \in X$. We must first check that $\varphi(a) \in S_X$ for each $a \in G$, i.e. that the function $x \mapsto a \cdot x$ is a bijection $X \rightarrow X$.

- *Injective:* Suppose $a \cdot x = a \cdot y$. Then $x = a^{-1} \cdot (a \cdot x) = a^{-1} \cdot (a \cdot y) = y$.
- *Surjective:* For any $x \in X$, we have $a \cdot (a^{-1} \cdot x) = x$, so $x \in \operatorname{im}(\varphi(a))$.

Hence $\varphi(a) \in S_X$. To see φ is a homomorphism, let $a, b \in G$. Then for all $x \in X$,

$$(\varphi(ab))(x) = (ab) \cdot x = a \cdot (b \cdot x) = (\varphi(a))((\varphi(b))(x)) = (\varphi(a)\varphi(b))(x).$$

(ii) Define $a \cdot x := (\varphi(a))(x)$. Then $(ab) \cdot x = (\varphi(ab))(x) = (\varphi(a)\varphi(b))(x) = a \cdot (b \cdot x)$, and $e \cdot x = (\varphi(e))(x) = x$. Hence X is a G -set. $\square$

Theorem 8 (1.11 — Cayley's Theorem). *Every group G is isomorphic to a subgroup of the symmetric group S_G .*

Proof. By Theorem 1.10(i), left translation $a \cdot x := ax$ gives a homomorphism $\varphi : G \rightarrow S_G$. If $a \in \ker \varphi$, then $\varphi(a) = \text{id}_G$, so $ax = x$ for all $x \in G$, giving $a = e$. Hence $\ker \varphi = \{e\}$, and by the First Isomorphism Theorem $G \cong G/\ker \varphi \cong \text{im } \varphi \leq S_G$. $\square$

Definition 1 (Normal core). *Let G be a group and $H \leq G$. The normal core of H in G is*

$$\text{core}(H) := \bigcap_{x \in G} xHx^{-1}.$$

It is the largest normal subgroup of G contained in H .

Theorem 9 (1.12). *Let G be a group and $H \leq G$ a subgroup of index n . Then there is a homomorphism $\varphi : G \rightarrow S_n$ such that $\ker \varphi = \text{core}(H)$.*

Proof. Since $|G/H| = n$, we have $S_{G/H} \cong S_n$. By Theorem 1.10(i), the action of G on the coset space G/H by left multiplication gives a homomorphism $\varphi : G \rightarrow S_n$ with $\varphi(a)(xH) = axH$. Then

$$\begin{aligned} \ker \varphi &= \{a \in G \mid axH = xH \text{ for all } x \in G\} \\ &= \{a \in G \mid x^{-1}ax \in H \text{ for all } x \in G\} \\ &= \{a \in G \mid a \in xHx^{-1} \text{ for all } x \in G\} \\ &= \bigcap_{x \in G} xHx^{-1} = \text{core}(H). \end{aligned} \quad \square$$

Corollary 1 (1.13). *Let G be a group with a normal subgroup H of index n . Then G/H is isomorphic to a subgroup of S_n .*

Proof. Since $H \trianglelefteq G$, conjugation fixes H , so $\text{core}(H) = \bigcap_{x \in G} xHx^{-1} = H$. By Theorem 1.12, $\ker \varphi = H$, and by the First Isomorphism Theorem $G/H \cong \text{im } \varphi \leq S_n$. $\square$

Theorem 10 (1.24 — Orbit-Stabiliser Theorem). *Let G be a group acting on a set X . Then the set of all orbits in X under G is a partition of X . Furthermore, for each $x \in X$ there is a bijection $Gx \rightarrow G/G_x$, and hence*

$$|Gx| = |G : G_x|.$$

Therefore, if X is finite,

$$|X| = \sum_{x \in C} |G : G_x|,$$

where $C \subseteq X$ contains exactly one element from each orbit.

Proof. (Partition.) Define $x \sim y$ if $x = ay$ for some $a \in G$. This is an equivalence relation: reflexivity follows from $e \cdot x = x$, symmetry from $x = ay \Rightarrow y = a^{-1}x$, and transitivity from $x = ay, y = bz \Rightarrow x = (ab)z$. The equivalence class of x is exactly the orbit Gx , so the orbits partition X .

(*Bijection* $Gx \rightarrow G/G_x$.) Define

$$\varphi : Gx \rightarrow G/G_x, \quad ax \mapsto aG_x.$$

The key observation is the following chain of equivalences, for any $a, b \in G$:

$$ax = bx \iff a^{-1}bx = x \iff a^{-1}b \in G_x \iff aG_x = bG_x.$$

- *Well-defined:* If $ax = bx$ (the same orbit element written using two different group elements), then $aG_x = bG_x$, so the coset φ assigns does not depend on the choice of representative.
- *Injective:* If $\varphi(ax) = \varphi(bx)$, i.e. $aG_x = bG_x$, then $ax = bx$ by the chain above.
- *Surjective:* Take any coset $aG_x \in G/G_x$. Since $a \in G$, we have $ax \in Gx$, and $\varphi(ax) = aG_x$.

Hence φ is a bijection, giving $|Gx| = |G/G_x| = |G : G_x|$. Summing over one representative per orbit gives $|X| = \sum_{x \in C} |G : G_x|$. $\square$

Theorem 11 (1.25 — Class Equation). *Let G be a finite group. Then the conjugacy classes partition G , and*

$$|G| = \sum_{a \in C} |G : C_G(a)|,$$

where $C \subseteq G$ contains exactly one element from each conjugacy class.

Proof. Let G act on itself by conjugation, i.e. $a \cdot x = axa^{-1}$. The orbit of x under this action is its conjugacy class $\text{ccl}(x) = \{axa^{-1} \mid a \in G\}$, and the stabiliser of x is $G_x = \{a \in G \mid axa^{-1} = x\} = C_G(x)$. Applying Theorem 1.24 directly gives the partition into orbits and $|G| = \sum_{a \in C} |G : C_G(a)|$. $\square$

Lecture 2

Theorem 12 (2.1). *Let G be a finite group of order p^n , where p is prime and $n \in \mathbb{N}$. Then:*

- (i) G has a non-trivial centre $Z(G)$;
- (ii) $Z(G) \cap K$ is non-trivial for any non-trivial normal subgroup K of G ;
- (iii) if H is a proper subgroup of G , then $H \subsetneq N_G(H)$. In particular, if $|H| = p^{n-1}$ then $H \trianglelefteq G$.

Proof. (i) Since $x \in Z(G)$ iff $\text{ccl}(x) = \{x\}$, the class equation refines to

$$p^n = |Z(G)| + \sum_{x \in C} |G : C_G(x)|,$$

where C runs over conjugacy classes with more than one element. For each $x \in C$, $|C_G(x)|$ properly divides p^n , so $p \nmid |G : C_G(x)|$. Hence p divides the sum, and therefore $p \mid |Z(G)|$. Since $e \in Z(G)$, we have $|Z(G)| \geq 1$, so in fact $|Z(G)| \geq p > 1$.

(ii) Intersecting the class decomposition of G with K gives $|K| = |Z(G) \cap K| + \sum_{x \in C} |\text{ccl}(x) \cap K|$. Since $K \trianglelefteq G$, each conjugacy class is either disjoint from K or contained in K , so $p \mid \sum |\text{ccl}(x) \cap K|$. As $p \mid |K|$, we get $p \mid |Z(G) \cap K|$, and the same argument as (i) gives $Z(G) \cap K$ non-trivial.

(iii) Let M be a maximal normal subgroup of G contained in H . Then G/M is a p -group, so by (i) it has a non-trivial centre $L/M \trianglelefteq G/M$. By the Correspondence Theorem, $L \trianglelefteq G$. Since $M \subsetneq L$ strictly, maximality of M forces $L \not\subseteq H$.

It remains to show $L \subseteq N_G(H)$. To show $\ell \in N_G(H)$ we need $\ell^{-1}h\ell \in H$ for all $h \in H$. Take $h \in H$ and $\ell \in L$. Since $\ell M \in L/M = Z(G/M)$, it commutes with every coset, so $(\ell M)(hM) = (hM)(\ell M)$, i.e. $\ell hM = h\ell M$. This gives $\ell^{-1}h\ell \in hM$. Since $h \in H$ and $M \subseteq H$, we have $hM \subseteq H$, so $\ell^{-1}h\ell \in H$. Hence $\ell \in N_G(H)$, so $L \subseteq N_G(H)$.

Since $L \subseteq N_G(H)$ but $L \not\subseteq H$, we have $H \subsetneq N_G(H)$. Finally, if $|H| = p^{n-1}$ then $N_G(H)$ has order strictly greater than p^{n-1} , and the only possibility is $p^n = |G|$, so $N_G(H) = G$ and $H \trianglelefteq G$. $\square$

Theorem 13 (2.2 — Burnside's Lemma). *Let G be a finite group acting on a finite set X , and for $g \in G$ let $X_g := \{x \in X \mid gx = x\}$. Then the number k of orbits in X under G is*

$$k = \frac{1}{|G|} \sum_{g \in G} |X_g|.$$

Proof. Let $S := \{(g, x) \in G \times X \mid gx = x\}$. Fixing g gives $|S| = \sum_{x \in X} |X_g|$; fixing x gives $|S| = \sum_{g \in G} |G_x|$. Let $C \subseteq X$ contain one element from each orbit. By the Orbit-Stabiliser Theorem, $|G_x| = |G|/|Gx| = |G|/|Ga|$ for all $x \in Ga$. Hence

$$\frac{1}{|G|} \sum_{g \in G} |X_g| = \frac{|S|}{|G|} = \frac{1}{|G|} \sum_{x \in X} |G_x| = \frac{1}{|G|} \sum_{a \in C} \sum_{x \in Ga} \frac{|G|}{|Ga|} = \frac{1}{|G|} \sum_{a \in C} |G| = k. \quad \square$$

Remark 1 (Pólya counting). *Burnside's Lemma counts distinct objects under symmetry. As an illustration: how many necklaces with p beads (p prime) can be made from n colours, where two necklaces are the same if one is a rotation of the other?*

Let X be the set of all colourings, so $|X| = n^p$. The cyclic group $G = \langle a \rangle$ of order p acts on X by rotation. For the identity, $X_e = X$ so $|X_e| = n^p$. For any $g \neq e$: since $|G|$ is prime, g generates G , so a necklace fixed by g must be fixed by all rotations, meaning all beads have the same colour. Hence $|X_g| = n$. Burnside then gives

$$k = \frac{1}{p} (n^p + (p-1)n).$$

Lecture 3

Definition 2 (p -group, Sylow p -subgroup). *Let p be a prime. A group G is a p -group if every element of G has order a power of p . If G is finite and $p^m \mid |G|$ but $p^{m+1} \nmid |G|$, then any subgroup of G of order p^m is called a Sylow p -subgroup of G .*

Theorem 14 (3.5 — First Sylow Theorem). *Let G be a finite group and p a prime. If $p^m \mid |G|$, then G has a subgroup of order p^m .*

Proof. If $m = 0$ the result is trivial. We proceed by induction on $n = |G|$, the case $n = 1$ being trivial. Assume the result holds for all groups of order less than n .

Case 1: $p \mid |Z(G)|$. By the Fundamental Theorem of Finitely Generated Abelian Groups, $Z(G)$ contains an element a of order p . Let $C = \langle a \rangle$. Since $a \in Z(G)$, we have $C \trianglelefteq G$, and $|G/C| = n/p$, which is divisible by p^{m-1} . By the induction hypothesis, G/C has a subgroup of order p^{m-1} , which by the Correspondence Theorem is of the form H/C for some $H \leq G$. Then $|H| = |H/C| \cdot |C| = p^{m-1} \cdot p = p^m$.

Case 2: $p \nmid |Z(G)|$. From the class equation $n = |Z(G)| + \sum_a |G : C_G(a)|$, since $p \mid n$ but $p \nmid |Z(G)|$, there must exist some $a \notin Z(G)$ with $p \nmid |G : C_G(a)|$. Since $|G| = |G : C_G(a)| \cdot |C_G(a)|$ and $p^m \mid |G|$ but $p \nmid |G : C_G(a)|$, it follows that $p^m \mid |C_G(a)|$. Since $a \notin Z(G)$, we have $C_G(a) \neq G$, so $|C_G(a)| < n$. By the induction hypothesis $C_G(a)$ contains a subgroup of order p^m , which is also a subgroup of G . $\square$

Corollary 2 (3.6 — Cauchy's Theorem). *If p is a prime dividing $|G|$, then G has an element of order p .*

Proof. By the First Sylow Theorem, G has a subgroup H of order p . Every non-identity element of H has order p . $\square$

Corollary 3 (3.7). *A finite group G is a p -group if and only if $|G|$ is a power of p .*

Proof. If $|G|$ is a power of p , then by Lagrange every element order divides $|G|$, hence is a power of p . Conversely, if $|G|$ is not a power of p , then some prime $q \neq p$ divides $|G|$, and by Cauchy's Theorem G has an element of order q , contradicting that G is a p -group. $\square$

Theorem 15 (3.8 — Second Sylow Theorem). *Let G be a finite group and p a prime. All Sylow p -subgroups of G are conjugate.*

Proof. Let $S, H \in \text{Syl}_p(G)$. Let S act on the coset space G/H by left translation: $s \cdot gH = sgH$.

Orbits have p -power size. By the Orbit-Stabiliser Theorem, each orbit has size $|S : S_{gH}|$, which divides $|S| = p^m$, hence is a power of p .

There must be a fixed point. Since H is a Sylow p -subgroup, $p \nmid |G|/|H| = |G/H|$. Writing $|G/H|$ as a sum of orbit sizes, each of which is a power of p , the only way this sum is not divisible by p is if at least one orbit has size $p^0 = 1$.

A fixed point gives conjugacy. If gH is a fixed point, then $s \cdot gH = gH$ for all $s \in S$, so $g^{-1}sg \in H$ for all $s \in S$. Hence $g^{-1}Sg \subseteq H$, and since $|g^{-1}Sg| = |S| = |H|$, we get $g^{-1}Sg = H$. So S and H are conjugate. $\square$

Corollary 4 (3.10). *A Sylow p -subgroup of G is unique if and only if it is normal in G .*

Proof. If P is the unique Sylow p -subgroup, then $gPg^{-1} = P$ for all $g \in G$ (since conjugates are also Sylow p -subgroups), so $P \trianglelefteq G$. Conversely, if $P \trianglelefteq G$ then P is its own conjugate, so by the Second Sylow Theorem it is the only Sylow p -subgroup. $\square$

Theorem 16 (3.9 — Third Sylow Theorem). *Let G be a finite group and p a prime. The number n_p of Sylow p -subgroups satisfies $n_p \mid |G|$ and $n_p \equiv 1 \pmod{p}$.*

Proof. $n_p \mid |G|$. By the Second Sylow Theorem, all Sylow p -subgroups are conjugate to a fixed $K \in \text{Syl}_p(G)$, so $n_p = |\text{ccl}(K)| = |G : N_G(K)|$, which divides $|G|$.

$n_p \equiv 1 \pmod{p}$. Let $P \in \text{Syl}_p(G)$. Since conjugates of Sylow p -subgroups are again Sylow p -subgroups (by the Second Sylow Theorem), P acts on the set $\text{Syl}_p(G)$ by conjugation:

$$x \cdot Q := xQx^{-1}, \quad x \in P, Q \in \text{Syl}_p(G).$$

This action partitions $\text{Syl}_p(G)$ into orbits. Let $F \subseteq \text{Syl}_p(G)$ denote the set of fixed points (those Q with $xQx^{-1} = Q$ for all $x \in P$), and let $\mathcal{O}$ denote the collection of orbits of size greater than 1. Then:

$$n_p = |\text{Syl}_p(G)| = |F| + \sum_{O \in \mathcal{O}} |O|.$$

By the Orbit-Stabiliser Theorem each $|O|$ divides $|P| = p^m$, so every $O \in \mathcal{O}$ has size p^e for some $e \geq 1$, hence $p \mid |O|$. Therefore $n_p \equiv |F| \pmod{p}$.

It remains to show $F = \{P\}$, which gives $n_p \equiv 1 \pmod{p}$. Clearly $P \in F$. If $Q \in F$, then $xQx^{-1} = Q$ for all $x \in P$, so $P \leq N_G(Q)$. By Corollary 3.10, since $Q \trianglelefteq N_G(Q)$, Q is the unique Sylow p -subgroup of $N_G(Q)$. But P is also a Sylow p -subgroup of $N_G(Q)$, so $P = Q$. Hence $F = \{P\}$. $\square$

Lecture 4 — Modules: the Basics

Definition 3 (4.1 — Left R -module). *Let R be a ring. An abelian group M is a left R -module if there is a map $R \times M \rightarrow M$, $(r, m) \mapsto rm$, such that for all $r, r_1, r_2 \in R$ and $m, m_1, m_2 \in M$:*

$$(i) \quad r(m_1 + m_2) = rm_1 + rm_2;$$

$$(ii) \quad (r_1 + r_2)m = r_1m + r_2m;$$

$$(iii) \quad (r_1r_2)m = r_1(r_2m);$$

$$(iv) \quad 1m = m \text{ if } 1 \in R.$$

Lemma 1 (4.7). *Let M be an R -module. Then for $a \in R$ and $m \in M$:*

$$(i) \quad 0m = 0;$$

$$(ii) \quad a0 = 0;$$

$$(iii) \quad (-a)m = -(am) = a(-m).$$

Proof. (i) $0m = (0 + 0)m = 0m + 0m$. Subtracting $0m$ from both sides gives $0 = 0m$.

(ii) $a0 = a(0 + 0) = a0 + a0$. Subtracting $a0$ from both sides gives $0 = a0$.

(iii) By (i) and axiom (ii): $0 = 0m = (a + (-a))m = am + (-a)m$, so $(-a)m = -(am)$. By (ii) and axiom (i): $0 = a0 = a(m + (-m)) = am + a(-m)$, so $a(-m) = -(am)$. $\square$

Definition 4 (4.8 — Submodule). *A subset N of an R -module M is an R -submodule of M if:*

$$(i) \quad N \neq \emptyset;$$

$$(ii) \quad a - b \in N \text{ for all } a, b \in N;$$

$$(iii) \quad ra \in N \text{ for all } a \in N, r \in R.$$

Theorem 17 (4.12). *Let $(N_i)_{i \in I}$ be a family of R -submodules of an R -module M . Then $\bigcap_{i \in I} N_i$ is also an R -submodule of M .*

Proof. Since $0 \in N_i$ for all i , we have $0 \in \bigcap N_i$, so it is non-empty. Let $x, y \in \bigcap N_i$ and $r \in R$. For each i , since N_i is a submodule, $x - y \in N_i$ and $rx \in N_i$. Hence $x - y$ and rx lie in $\bigcap N_i$. $\square$

Definition 5 (4.13 — Generated submodule, cyclic module). *Let $S \subseteq M$. The family $\mathcal{A} = \{N \mid N \text{ is an } R\text{-submodule of } M \text{ containing } S\}$ is non-empty (since $M \in \mathcal{A}$), so by Theorem 4.12 the intersection $\bigcap_{N \in \mathcal{A}} N$ is an R -submodule. It is the smallest submodule of M containing S , called the submodule generated by S and denoted (S) . If $S = \{x_1, \dots, x_n\}$ we write $(x_1, \dots, x_n)$. An R -module M is cyclic if $M = (x)$ for some $x \in M$.*

Theorem 18 (4.15). *Let R be a ring with unity and $x_1, \dots, x_n \in M$. Then*

$$(x_1, \dots, x_n) = \sum_{i=1}^n Rx_i := \{r_1x_1 + \dots + r_nx_n \mid r_i \in R\}.$$

Proof. $(\supseteq)$ The submodule $(x_1, \dots, x_n)$ contains each x_i . By closure under scalar multiplication it contains rx_i for all $r \in R$, and by closure under addition it contains every sum $r_1x_1 + \dots + r_nx_n$. Hence $\sum Rx_i \subseteq (x_1, \dots, x_n)$.

$(\subseteq)$ We show $\sum Rx_i$ is a submodule containing each x_i ; then $(x_1, \dots, x_n) \subseteq \sum Rx_i$ follows from minimality of $(x_1, \dots, x_n)$.

Each $x_i \in \sum Rx_i$ since $x_i = 1 \cdot x_i$ (using unity). For the submodule conditions: $0 = \sum 0 \cdot x_i \in \sum Rx_i$, so it is non-empty. For $\sum r_ix_i, \sum s_ix_i \in \sum Rx_i$ and $r \in R$,

$$\sum r_ix_i - \sum s_ix_i = \sum (r_i - s_i)x_i \in \sum Rx_i, \quad r \sum r_ix_i = \sum (rr_i)x_i \in \sum Rx_i. \quad \square$$

Definition 6 (4.18 — Sum of submodules). *Let $N_1, \dots, N_k$ be R -submodules of M . Their sum $\sum_{i=1}^k N_i$ is the submodule of M generated by $\bigcup_{i=1}^k N_i$, i.e. the smallest submodule of M containing each N_i .*

Lecture 5 — Modules: continuation

Theorem 19 (4.19). *Let $N_1, \dots, N_k$ be R -submodules of an R -module M . Then*

$$\sum_{i=1}^k N_i = \{x_1 + \dots + x_k \mid x_i \in N_i\}.$$

Proof. Let

$$A = \{x_1 + \dots + x_k \mid x_i \in N_i\}.$$

It is straightforward to check that A is a submodule of M , and it contains every N_i . Hence, by minimality,

$$\sum_{i=1}^k N_i \subseteq A.$$

Conversely, $\sum_{i=1}^k N_i$ contains every N_i and is closed under addition, so it contains every element of A . Therefore

$$A \subseteq \sum_{i=1}^k N_i.$$

□

Definition 7 (4.21 — Direct sum). *The sum $N_1 + \cdots + N_k$ is direct if every element has a unique representation*

$$x = x_1 + \cdots + x_k, \quad x_i \in N_i.$$

In this case we write

$$N_1 \oplus \cdots \oplus N_k.$$

Theorem 20 (4.22). *Let $N_1, \dots, N_k$ be submodules of an R -module M . The following are equivalent:*

- (i) $\sum_{i=1}^k N_i$ is direct;
- (ii) if $\sum_{i=1}^k x_i = 0$, where $x_i \in N_i$, then every $x_i = 0$;
- (iii) for every i ,

$$N_i \cap \sum_{j \neq i} N_j = \{0\}.$$

Proof. (i) $\Rightarrow$ (ii): Zero already has the representation $0 + \cdots + 0$, so uniqueness implies that every $x_i = 0$.

(ii) $\Rightarrow$ (iii): If

$$x \in N_i \cap \sum_{j \neq i} N_j,$$

then $x = \sum_{j \neq i} x_j$ for some $x_j \in N_j$. Hence

$$x - \sum_{j \neq i} x_j = 0,$$

so (ii) gives $x = 0$.

(iii) $\Rightarrow$ (i): Suppose

$$\sum_{i=1}^k x_i = \sum_{i=1}^k y_i, \quad x_i, y_i \in N_i.$$

For every i ,

$$x_i - y_i = - \sum_{j \neq i} (x_j - y_j) \in N_i \cap \sum_{j \neq i} N_j = \{0\}.$$

Thus $x_i = y_i$ for every i , so the representation is unique.

□

Definition 8 (4.23 — *R*-homomorphism). Let M and N be *R*-modules. A map $f: M \rightarrow N$ is an *R*-homomorphism if, for all $x, y \in M$ and $r \in R$,

$$f(x + y) = f(x) + f(y), \quad f(rx) = rf(x).$$

The set of all *R*-homomorphisms $M \rightarrow N$ is denoted

$$\text{Hom}_R(M, N).$$

An *R*-homomorphism $M \rightarrow M$ is called an endomorphism, and

$$\text{End}_R(M) := \text{Hom}_R(M, M).$$

Theorem 21 (4.30). Let $M = \bigoplus_{i=1}^k M_i$ be a direct sum of *R*-modules. Then

$$\text{Hom}_R(M, M) \cong (\text{Hom}_R(M_j, M_i))_{1 \leq i, j \leq k}$$

as rings, where the right-hand side is a ring T of $k \times k$ matrices under the usual matrix addition and multiplication, with (i, j) entry in $\text{Hom}_R(M_j, M_i)$.

Proof. Let $\lambda_j : M_j \rightarrow M$ be the natural inclusion and $\pi_i : M \rightarrow M_i$ the natural projection. Two key identities hold:

$$\sum_{\ell=1}^k \lambda_\ell \pi_\ell = \text{id}_M, \quad \pi_p \lambda_q = \delta_{pq} \text{id}_{M_p}.$$

The first says every $x \in M$ equals the sum of its components; the second says projecting the q -th inclusion gives zero unless $p = q$.

Define $\sigma : \text{Hom}_R(M, M) \rightarrow T$ by $\sigma(\phi) = (\pi_i \phi \lambda_j)_{ij}$.

Ring homomorphism. Additivity is immediate. For multiplication, using the first identity:

$$(\sigma(\phi)\sigma(\psi))_{ij} = \sum_{\ell} \pi_i \phi \lambda_\ell \pi_\ell \psi \lambda_j = \pi_i \phi \left(\sum_{\ell} \lambda_\ell \pi_\ell \right) \psi \lambda_j = \pi_i \phi \text{id}_M \psi \lambda_j = (\sigma(\phi\psi))_{ij}.$$

Injective. Suppose $\sigma(\phi) = 0$, so $\pi_i \phi \lambda_j = 0$ for all i, j . Applying λ_i on the left and summing over i using the first identity:

$$\phi \lambda_j = \text{id}_M \cdot \phi \lambda_j = \left(\sum_i \lambda_i \pi_i \right) \phi \lambda_j = \sum_i \lambda_i (\pi_i \phi \lambda_j) = 0.$$

Now applying π_j on the right and summing over j :

$$\phi = \phi \cdot \text{id}_M = \phi \left(\sum_j \lambda_j \pi_j \right) = \sum_j (\phi \lambda_j) \pi_j = 0.$$

Surjective. Given $f = (f_{ij}) \in T$, set $\phi = \sum_{i,j} \lambda_i f_{ij} \pi_j$. Using the second identity:

$$(\sigma(\phi))_{st} = \pi_s \phi \lambda_t = \sum_{i,j} \pi_s \lambda_i f_{ij} \pi_j \lambda_t = \sum_{i,j} \delta_{si} f_{ij} \delta_{jt} = f_{st}.$$

Hence $\sigma(\phi) = f$, so σ is surjective. □

Definition 9 (Quotient module). Let N be an R -submodule of an R -module M . The set of cosets $M/N = \{a + N \mid a \in M\}$ becomes an R -module under

$$(a + N) + (b + N) = (a + b) + N, \quad r(a + N) = ra + N.$$

It is called the quotient module of M by N .

Theorem 22 (4.31 — Correspondence Theorem for Modules). Let N be an R -submodule of an R -module M . Every R -submodule of M/N is of the form U/N , where U is an R -submodule of M containing N .

Proof. The canonical projection $f : M \rightarrow M/N$, $f(x) = x + N$, is an R -homomorphism:

$$f(x+y) = (x+y)+N = (x+N)+(y+N) = f(x)+f(y), \quad f(rx) = rx+N = r(x+N) = rf(x).$$

Let X be an R -submodule of M/N . The structure of the proof is: define U as the preimage of X under f , show U is a submodule of M containing N , and conclude $X = U/N$.

Set $U = f^{-1}(X) = \{x \in M \mid f(x) \in X\}$.

$N \subseteq U$. For any $x \in N$, $f(x) = x + N = 0 + N = 0_{M/N} \in X$ (since X is a submodule, it contains 0). Hence $x \in U$.

U is an R -submodule. Since $0 \in N \subseteq U$, U is non-empty. For $x, y \in U$ and $r \in R$, since f is an R -homomorphism:

$$f(x - y) = f(x) - f(y) \in X, \quad f(rx) = rf(x) \in X,$$

so $x - y \in U$ and $rx \in U$.

$X = U/N$. By definition of U , $f(U) \subseteq X$. For the reverse inclusion, take any $x \in X$; since f is surjective there exists $y \in M$ with $f(y) = x$, so $y \in U$ and $x \in f(U)$. Hence $X = f(U) = U/N$. $\square$

Theorem 23 (4.32 — First Isomorphism Theorem for Modules). Let $f : M \rightarrow N$ be an R -homomorphism. Then $M/\ker f \cong f(M)$.

Proof. Define $g : M/\ker f \rightarrow f(M)$ by $g(x + \ker f) = f(x)$. We know from group theory that g is an isomorphism of abelian groups (well-defined and injective because $x + \ker f = y + \ker f \iff x - y \in \ker f \iff f(x) = f(y)$; surjective onto $f(M)$ by definition). It remains to check that g respects the R -action:

$$g(r(x + \ker f)) = g(rx + \ker f) = f(rx) = rf(x) = rg(x + \ker f). \quad \square$$

Theorem 24 (Second Isomorphism Theorem for Modules). Let U and V be R -submodules of an R -module M . Then $(U + V)/V \cong U/(U \cap V)$.

Proof. Define $f : U \rightarrow (U + V)/V$ by $f(u) = u + V$.

f is an R -homomorphism. For $u_1, u_2 \in U$ and $r \in R$:

$$f(u_1 + u_2) = (u_1 + u_2) + V = (u_1 + V) + (u_2 + V) = f(u_1) + f(u_2),$$

$$f(ru) = ru + V = r(u + V) = rf(u).$$

f is surjective. Any element of $(U + V)/V$ has the form $(u + v) + V = u + V = f(u)$ for some $u \in U$, $v \in V$ (since $v \in V$ means $v + V = V$).

$\ker f = U \cap V$. We have $f(u) = 0$ iff $u + V = V$ iff $u \in V$, so $\ker f = \{u \in U \mid u \in V\} = U \cap V$.

By the First Isomorphism Theorem, $U/(U \cap V) \cong (U + V)/V$. $\square$

Theorem 25 (Third Isomorphism Theorem for Modules). *Let $U \subseteq V$ be R -submodules of an R -module M . Then $(M/U)/(V/U) \cong M/V$.*

Proof. Define $f : M/U \rightarrow M/V$ by $f(x+U) = x+V$.

f is well-defined. If $x+U = y+U$ then $x-y \in U \subseteq V$, so $x+V = y+V$.

f is an R -homomorphism. For $x+U, y+U \in M/U$ and $r \in R$:

$$f((x+U)+(y+U)) = f((x+y)+U) = (x+y)+V = (x+V)+(y+V) = f(x+U)+f(y+U),$$

$$f(r(x+U)) = f(rx+U) = rx+V = r(x+V) = rf(x+U).$$

f is surjective. For any $x+V \in M/V$, $f(x+U) = x+V$.

$\ker f = V/U$. We have $f(x+U) = 0$ iff $x+V = V$ iff $x \in V$, so $\ker f = \{x+U \mid x \in V\} = V/U$.

By the First Isomorphism Theorem, $(M/U)/(V/U) \cong M/V$. $\square$

Theorem 26 (4.33). *Let $A \leq M$ and $B \leq N$ be R -submodules. Then $(M \times N)/(A \times B) \cong M/A \times N/B$.*

Proof. Define $f : M \times N \rightarrow M/A \times N/B$ by $f(m, n) = (m+A, n+B)$. This is a surjective R -homomorphism, and $\ker f = \{(m, n) \mid m \in A, n \in B\} = A \times B$. By the First Isomorphism Theorem, $(M \times N)/(A \times B) \cong M/A \times N/B$. $\square$

Lecture 6 — Simple, Completely Reducible, and Free Modules

Definition 10 (5.1 — Simple module). *Let R be a ring with unity. An R -module M is called simple (or irreducible) if $M \neq 0$ and the only R -submodules of M are 0 and M .*

Theorem 27 (5.5 — Characterisation of Simple Modules). *Let R be a ring with unity and M an R -module. The following are equivalent:*

- (i) M is simple;
- (ii) $M \neq 0$, and M is generated by any $0 \neq x \in M$;
- (iii) $M \cong R/I$, where I is a maximal left ideal of R .

Proof. (i) $\Rightarrow$ (ii): Let $0 \neq x \in M$. Then Rx is a non-zero submodule of M , so $Rx = M$ by simplicity.

(ii) $\Rightarrow$ (iii): Take $0 \neq x \in M$ and define $f : R \rightarrow M$ by $f(r) = rx$. This is a surjective R -homomorphism, so letting $I = \ker f$, the First Isomorphism Theorem gives $R/I \cong M$. Note that (ii) implies M is simple: if $0 \neq N \leq M$ and $0 \neq x \in N$, then $M = Rx \subseteq N$. Hence $R/I \cong M$ is simple. By the Correspondence Theorem (4.31), any left ideal J with $I \subseteq J \subseteq R$ corresponds to a submodule J/I of R/I . Simplicity forces $J/I = 0$ or $J/I = R/I$, i.e. $J = I$ or $J = R$, so I is maximal.

(iii) $\Rightarrow$ (i): Let $0 \neq N \leq M \cong R/I$. By the Correspondence Theorem (4.31), $N \cong J/I$ for some left ideal J with $I \subsetneq J \subseteq R$. Since I is maximal, $J = R$, so $N = M$. $\square$

Theorem 28 (5.6 — Schur's Lemma). *Let M be a simple R -module. Then $\text{Hom}_R(M, M)$ is a division ring.*

Proof. Let $0 \neq f \in \text{Hom}_R(M, M)$. We show f is invertible and $f^{-1} \in \text{Hom}_R(M, M)$.

f is bijective. Since $\ker f$ and $\text{im } f$ are submodules of M , simplicity forces each to be 0 or M . As $f \neq 0$, we have $\ker f \neq M$ and $\text{im } f \neq 0$, so $\ker f = 0$ (injective) and $\text{im } f = M$ (surjective).

f^{-1} is an R -homomorphism. We cannot push f^{-1} through operations directly, but we can push f through them. The strategy is: to prove $f^{-1}(u) = v$, it suffices to show $f(u) = f(v)$, since f is injective. For $x, y \in M$ and $r \in R$:

$$\begin{aligned} f(f^{-1}(x + y)) &= x + y = f(f^{-1}(x)) + f(f^{-1}(y)) = f(f^{-1}(x) + f^{-1}(y)), \\ f(f^{-1}(rx)) &= rx = rf(f^{-1}(x)) = f(rf^{-1}(x)), \end{aligned}$$

where we used that f is itself an R -homomorphism in the last step of each line. Injectivity of f gives $f^{-1}(x + y) = f^{-1}(x) + f^{-1}(y)$ and $f^{-1}(rx) = rf^{-1}(x)$. $\square$

Definition 11 (5.7 — Completely reducible module). Let R be a ring with unity. An R -module M is called completely reducible (or semisimple) if $M = \sum_{\alpha \in \Lambda} M_\alpha$, where each M_α is a simple R -submodule.

Theorem 29 (5.9). Let $M = \sum_{\alpha \in \Lambda} M_\alpha$ be a sum of simple R -submodules, and let K be a submodule of M . Then there exists $\Lambda' \subseteq \Lambda$ such that $M = K \oplus (\bigoplus_{\alpha \in \Lambda'} M_\alpha)$.

Proof. Let S be the family of subsets $A \subseteq \Lambda$ satisfying two conditions: $\bigoplus_{\alpha \in A} M_\alpha$ is a direct sum, and $K \cap \bigoplus_{\alpha \in A} M_\alpha = 0$. Together these mean $K \oplus \bigoplus_{\alpha \in A} M_\alpha$ is a genuine direct sum. Since $\emptyset$ satisfies both conditions trivially, $S \neq \emptyset$. Partially order S by inclusion; every chain (A_i) in S has upper bound $\bigcup A_i$ in S . By Zorn's Lemma, S has a maximal element A .

Set $N = K \oplus \bigoplus_{\alpha \in A} M_\alpha$. We claim $N = M$. Fix any $\beta \in \Lambda$. Since M_β is simple, $M_\beta \cap N$ is either 0 or M_β . If $M_\beta \cap N = 0$, then M_β intersects neither K nor $\bigoplus_{\alpha \in A} M_\alpha$, so $A \cup \{\beta\}$ would satisfy both conditions in S , contradicting maximality of A . Hence $M_\beta \cap N = M_\beta$, i.e. $M_\beta \subseteq N$, for every $\beta \in \Lambda$. Since $M = \sum_{\beta} M_\beta$, we conclude $N = M$. $\square$

Corollary 5 (5.10). If $M = \sum_{\alpha \in \Lambda} M_\alpha$ is a sum of simple R -submodules, then there exists $\Lambda' \subseteq \Lambda$ such that $M = \bigoplus_{\alpha \in \Lambda'} M_\alpha$.

Proof. Apply Theorem 5.9 with $K = \{0\}$. $\square$

Definition 12 (5.11 — Linear independence and basis). A sequence $x_1, \dots, x_n$ in an R -module M is linearly independent if $\sum_{i=1}^n a_i x_i = 0$ implies $a_1 = \dots = a_n = 0$ for any $a_i \in R$. A subset $S \subseteq M$ is linearly independent if every finite sequence of distinct elements of S is linearly independent.

A subset $B \subseteq M$ is a basis of M if B generates M and B is linearly independent. An R -module that has a basis is called a free module.

Lecture 7 — Free Modules and Chain Conditions

Theorem 30 (5.19). Let M be a free R -module with basis $\{f_1, \dots, f_n\}$. Then $M \cong R^n$ as R -modules.

Proof. Define $\varphi : M \rightarrow R^n$ by $\varphi(\sum_{i=1}^n r_i f_i) = (r_1, \dots, r_n)$, i.e. read off the coordinates in the basis. Since every element of M has a unique representation as $\sum r_i f_i$ (by linear independence of the f_i), φ is well-defined. It is clearly an R -homomorphism, injective (if $(r_1, \dots, r_n) = 0$ then each $r_i = 0$, so $m = 0$), and surjective (every $(r_1, \dots, r_n) \in R^n$ is the image of $\sum r_i f_i$). $\square$

Theorem 31 (5.20). *Let M be a finitely generated free R -module. Then every basis of M is finite.*

Proof. Let $\{e_i\}_{i \in \Lambda}$ be a basis and $\{x_1, \dots, x_n\}$ a finite generating set. Since elements of a module are finite sums by definition, each x_j involves only finitely many nonzero coefficients, so is supported on a finite subset $S_j \subseteq \Lambda$. Every basis element must appear in the expression of some generator (otherwise it could not be reached), so $\Lambda \subseteq S_1 \cup \dots \cup S_n$, a finite union of finite sets. Hence Λ is finite. $\square$

Theorem 32 (5.21). *Let M be a finitely generated free module over a commutative ring R . Then all bases of M have the same number of elements.*

Proof. Suppose M has two bases of sizes m and n respectively. By Theorem 5.19, $M \cong R^m$ and $M \cong R^n$, so $R^m \cong R^n$. It suffices to show $m = n$.

Assume for contradiction $m < n$. Let $\phi : R^m \rightarrow R^n$ be an R -isomorphism with inverse $\psi = \phi^{-1}$. Let $(e_1, \dots, e_m)$ and $(f_1, \dots, f_n)$ be the standard bases of R^m and R^n . Since ϕ is an R -homomorphism, it is determined by where it sends each e_i , and each image lands in R^n , so we can write

$$\phi(e_i) = \sum_{j=1}^n a_{ji} f_j, \quad \psi(f_j) = \sum_{k=1}^m b_{kj} e_k.$$

Let $A = (a_{ji})$ be the resulting $n \times m$ matrix and $B = (b_{kj})$ the $m \times n$ matrix. Since $\psi\phi = \text{id}_{R^m}$ and $\phi\psi = \text{id}_{R^n}$, substituting the expressions above and using linear independence of the bases gives $BA = I_m$ and $AB = I_n$.

Pad these to $n \times n$ matrices by setting $A' = \begin{pmatrix} A & 0 \end{pmatrix}$ and $B' = \begin{pmatrix} B \\ 0 \end{pmatrix}$. Then $A'B' = I_n$, so $\det(A'B') = 1$. But $B'A' = \begin{pmatrix} I_m & 0 \\ 0 & 0 \end{pmatrix}$, so $\det(B'A') = 0$. Since R is commutative, determinants multiply, giving

$$1 = \det(A'B') = \det(A') \det(B') = \det(B'A') = 0,$$

a contradiction. By symmetry $n < m$ also fails, so $m = n$. $\square$

Definition 13 (5.22 — Rank). *The number of elements in any basis of a finitely generated free module M over a commutative ring R is called the rank of M , written $\text{rk}(M)$. If R is a field, the rank is called the dimension.*

Lemma 2 (5.24). *Every finitely generated R -module is a homomorphic image of a finitely generated free module.*

Proof. Let M be generated by $x_1, \dots, x_n$. Define $\phi : R^n \rightarrow M$ by $\phi(r_1, \dots, r_n) = \sum_{i=1}^n r_i x_i$. This is a surjective R -homomorphism (surjective since the x_i generate M). The generators x_i may satisfy relations, so ϕ need not be injective — but by the First Isomorphism Theorem $M \cong R^n / \ker \phi$, expressing M as a quotient of the free module R^n . $\square$

Lemma 3 (5.25). *Let R be a PID and F a free R -module of rank n . Then every submodule $K \leq F$ is free of rank $m \leq n$.*

Proof. By Theorem 5.19, $F \cong R^n$, so we work with $K \leq R^n$ and induct on n . The base case $n = 0$ is trivial since $K = 0$ is free on the empty set.

Define $\pi : R^n \rightarrow R$ by $\pi(x_1, \dots, x_n) = x_1$, and restrict it to K .

Case 1: $\pi|_K = 0$. Every element of K has first coordinate zero, so $K \leq \{0\} \times R^{n-1} \cong R^{n-1}$. By induction K is free of rank $\leq n - 1 \leq n$.

Case 2: $\pi|_K \neq 0$. The image $\pi(K)$ is a nonzero ideal of R . Since R is a PID, $\pi(K) = Ra$ for some $a \neq 0$. Pick $k \in K$ with $\pi(k) = a$.

We claim $K = Rk \oplus \ker(\pi|_K)$, where $\ker(\pi|_K) = \{x \in K \mid x_1 = 0\}$ is the set of elements of K with first coordinate zero, so $\ker(\pi|_K) \leq \{0\} \times R^{n-1} \cong R^{n-1}$.

$K = Rk + \ker(\pi|_K)$: For any $x \in K$, write $\pi(x) = ba$ for some $b \in R$. Then $\pi(x - bk) = ba - ba = 0$, so $x - bk \in \ker(\pi|_K)$, and $x = bk + (x - bk)$.

The sum is direct: If $ck \in \ker(\pi|_K)$, then $0 = \pi(ck) = ca$. Since R is a PID (hence an integral domain) and $a \neq 0$, we get $c = 0$.

By induction $\ker(\pi|_K)$ is free of rank $s \leq n - 1$, and $Rk \cong R$ is free of rank 1. Hence $K \cong R^s \oplus R = R^{s+1}$, which has rank $m = s + 1 \leq n$. $\square$

Lecture 8 — Noetherian and Artinian Modules

Definition 14 (6.1 — Noetherian and Artinian modules). *An R -module M is called Noetherian if every ascending chain of submodules*

$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \dots$$

stabilises, i.e. there exists k such that $M_k = M_{k+1} = \dots$. This is called the ascending chain condition (ACC). An R -module M is called Artinian if every descending chain of submodules stabilises (descending chain condition, DCC).

Theorem 33 (6.5). *Let M be an R -module. The following are equivalent:*

- (i) M is Noetherian;
- (ii) every submodule of M is finitely generated;
- (iii) every non-empty family S of submodules of M has a maximal element.

Proof. (i) $\Rightarrow$ (ii): Let $N \leq M$ and suppose N is not finitely generated. For any finite choice $a_1, \dots, a_k \in N$ we have $N \neq (a_1, \dots, a_k)$, so we can always pick $a_{k+1} \in N \setminus (a_1, \dots, a_k)$. This produces an infinite strictly ascending chain $(a_1) \subsetneq (a_1, a_2) \subsetneq \dots$, contradicting that M is Noetherian.

(ii) $\Rightarrow$ (iii): Suppose S has no maximal element. Then starting from any $N_0 \in S$ we can always find a strictly larger member, producing an infinite strictly ascending chain $N_0 \subsetneq N_1 \subsetneq N_2 \subsetneq \dots$. Let $N = \bigcup_i N_i$. For $x, y \in N$ and $r \in R$, both x and y lie in some N_ℓ (take the larger index), so $x - y, rx \in N_\ell \subseteq N$; hence N is a submodule. By (ii), $N = (a_1, \dots, a_n)$ for some $a_j \in N$. Each a_j lies in some N_{u_j} , so all generators lie in N_μ for $\mu = \max\{u_1, \dots, u_n\}$. But then $N = (a_1, \dots, a_n) \subseteq N_\mu \subseteq N$, so $N_\mu = N$, contradicting that the chain is strictly ascending.

(iii) $\Rightarrow$ (i): Given an ascending chain $M_1 \subseteq M_2 \subseteq \dots$, by (iii) the family $\{M_i\}$ has a maximal element M_k . Then $M_k = M_{k+1} = \dots$, so the chain stabilises. $\square$

Definition 15 (6.6 — Finitely cogenerated). An R -module M is finitely cogenerated if for every family $(M_\alpha)_{\alpha \in \Lambda}$ of submodules of M ,

$$\bigcap_{\alpha \in \Lambda} M_\alpha = 0 \implies \bigcap_{\alpha \in \Lambda'} M_\alpha = 0$$

for some finite subset $\Lambda' \subseteq \Lambda$.

Theorem 34 (6.8). Let M be an R -module. The following are equivalent:

- (i) M is Artinian;
- (ii) every quotient module of M is finitely cogenerated;
- (iii) every non-empty family S of submodules of M has a minimal element.

Proof. (i) $\Rightarrow$ (ii): Let $N \leq M$ and take any family $(M_\alpha/N)_{\alpha \in \Lambda}$ of submodules of M/N with $\bigcap_{\alpha \in \Lambda} M_\alpha/N = 0$, i.e. $\bigcap_{\alpha \in \Lambda} M_\alpha = N$. Consider the family of finite intersections $\{\bigcap_{\alpha \in \Lambda'} M_\alpha \mid \Lambda' \subseteq \Lambda \text{ finite}\}$. As we add more submodules to the intersection it can only get smaller, so this is a descending family of submodules of M . Since M is Artinian every descending chain stabilises, so there exists a finite $\Lambda' \subseteq \Lambda$ such that

$$\bigcap_{\alpha \in \Lambda'} M_\alpha = \bigcap_{\alpha \in \Lambda} M_\alpha = N.$$

Hence $\bigcap_{\alpha \in \Lambda'} M_\alpha/N = 0$, so M/N is finitely cogenerated.

(ii) $\Rightarrow$ (iii): Suppose S has no minimal element. Then starting from any $N_0 \in S$ we can always find a strictly smaller member, producing an infinite strictly descending chain $N_0 \supsetneq N_1 \supsetneq N_2 \supsetneq \dots$. Let $N = \bigcap_i N_i$, which is a submodule of M . By (ii), M/N is finitely cogenerated. The family (N_i/N) of submodules of M/N satisfies $\bigcap_i N_i/N = N/N = 0$, so by finite cogeneratedness there exists a finite subset, say $\{N_0, \dots, N_k\}$, with $\bigcap_{i=0}^k N_i/N = 0$, i.e. $\bigcap_{i=0}^k N_i = N$. Since the chain is descending, $\bigcap_{i=0}^k N_i = N_k$, so $N_k = N \subseteq N_{k+1} \subsetneq N_k$, a contradiction.

(iii) $\Rightarrow$ (i): Given a descending chain $M_1 \supseteq M_2 \supseteq \dots$, by (iii) the family $\{M_i\}$ has a minimal element M_k . Then $M_k = M_{k+1} = \dots$, so the chain stabilises. $\square$

Theorem 35 (6.9). Let M be a Noetherian (respectively Artinian) R -module and $K \leq M$ a submodule. Then K and M/K are both Noetherian (respectively Artinian).

Proof. Noetherian case. By Theorem 6.5, M is Noetherian if and only if every submodule of M is finitely generated. Every submodule of K is also a submodule of M , hence finitely generated. By Theorem 6.5 again, K is Noetherian. For M/K : by Theorem 4.31 (Correspondence Theorem), every submodule of M/K is of the form L/K for some submodule L with $K \subseteq L \subseteq M$. Since L is a submodule of M , it is finitely generated, and hence L/K is finitely generated. By Theorem 6.5, M/K is Noetherian.

Artinian case. Any descending chain $K \supseteq L_1 \supseteq L_2 \supseteq \dots$ of submodules of K is also a descending chain in M , so it stabilises. Hence K is Artinian. For M/K : a descending chain $M/K \supseteq L_1/K \supseteq L_2/K \supseteq \dots$ of submodules of M/K corresponds via Theorem 4.31 to a descending chain $M \supseteq L_1 \supseteq L_2 \supseteq \dots$ in M , which stabilises. Hence M/K is Artinian. $\square$

Theorem 36 (6.10). *Let M be an R -module and $N \leq M$ a submodule. Then M is Noetherian (respectively Artinian) if and only if both N and M/N are Noetherian (respectively Artinian).*

Proof. ($\Rightarrow$) This is Theorem 6.9.

($\Leftarrow$) *Noetherian case.* Assume N and M/N are Noetherian. Let $K \leq M$ be any submodule; we show K is finitely generated, which by Theorem 6.5 gives that M is Noetherian.

By the Second Isomorphism Theorem, $K/(N \cap K) \cong (K + N)/N$. Since $(K + N)/N$ is a submodule of M/N and M/N is Noetherian, it is finitely generated by Theorem 6.5, and hence so is $K/(N \cap K)$. Say $K/(N \cap K)$ is generated by $\bar{x}_1, \dots, \bar{x}_m$ with $x_i \in K$. Then

$$K = (x_1, \dots, x_m) + (N \cap K).$$

Since $N \cap K$ is a submodule of N and N is Noetherian, $N \cap K$ is finitely generated by Theorem 6.5, say by $y_1, \dots, y_n \in N \cap K$. Therefore

$$K = (x_1, \dots, x_m, y_1, \dots, y_n),$$

so K is finitely generated and M is Noetherian.

Artinian case. Assume N and M/N are Artinian. Let $M \supseteq K_1 \supseteq K_2 \supseteq \dots$ be a descending chain in M . The induced chains

$$K_1 \cap N \supseteq K_2 \cap N \supseteq \dots \quad \text{in } N, \quad \frac{K_1 + N}{N} \supseteq \frac{K_2 + N}{N} \supseteq \dots \quad \text{in } M/N$$

both stabilise (since N and M/N are Artinian), say at indices n and m respectively. Set $q = \max\{m, n\}$. For $i \geq q$ we have $K_i \cap N = K_q \cap N$ and $K_i + N = K_q + N$. One can then show using the Second Isomorphism Theorem that $K_i = K_q$, so the original chain stabilises and M is Artinian. $\square$

Corollary 6 (6.11). *If $M_1, \dots, M_n$ are Noetherian (respectively Artinian) R -modules, then $M_1 \oplus \dots \oplus M_n$ is Noetherian (respectively Artinian).*

Proof. By induction it suffices to treat the case of two modules M and N . By Theorem 6.10, to show $M \oplus N$ is Noetherian (respectively Artinian) it suffices to find a single submodule of $M \oplus N$ such that both the submodule and its quotient module are Noetherian (respectively Artinian). Take $M \oplus 0 \leq M \oplus N$. Then:

- the submodule $M \oplus 0 \cong M$ is Noetherian (respectively Artinian) by assumption;
- the quotient $(M \oplus N)/(M \oplus 0) \cong N$ is Noetherian (respectively Artinian) by assumption.

Hence by Theorem 6.10, $M \oplus N$ is Noetherian (respectively Artinian). $\square$

Definition 16 (6.12 — Noetherian and Artinian rings). *A ring R is called left Noetherian (respectively left Artinian) if R , regarded as a left R -module, is Noetherian (respectively Artinian). By default, Noetherian (respectively Artinian) ring means left Noetherian (respectively left Artinian).*

Theorem 37 (6.13). *Let R be a ring. The following are equivalent:*

- (i) R is Noetherian (respectively Artinian);
- (ii) every left ideal A of R is finitely generated (respectively every R/A is finitely cogenerated);
- (iii) every non-empty set of left ideals of R has a maximal (respectively minimal) element.

Proof. This is Theorem 6.5 (respectively 6.8) applied to R regarded as a left R -module, where submodules of R are precisely the left ideals of R . $\square$

Theorem 38 (6.19). *Let $R_1, \dots, R_n$ be Noetherian (respectively Artinian) rings. Then $R_1 \oplus \dots \oplus R_n$ is Noetherian (respectively Artinian).*

Proof. This follows immediately from Corollary 6.11, viewing each R_i as an R -module. $\square$

Product of ideals. For ideals A and B in a ring R , define

$$AB = \left\{ \sum_{i=1}^n a_i b_i \mid n \in \mathbb{N}, a_i \in A, b_i \in B \right\}.$$

Then AB is again an ideal of R .

Definition 17 (6.20 — Prime ideal). *Let R be a ring. An ideal $P \neq R$ is prime if whenever A and B are ideals in R with $AB \subseteq P$, then $A \subseteq P$ or $B \subseteq P$.*

(For commutative R this recovers the classical definition: $ab \in P \Rightarrow a \in P$ or $b \in P$.)

Theorem 39 (6.21). *Let R be a ring with unity. Then every maximal ideal is prime.*

Proof. Let M be a maximal ideal and let A, B be ideals with $AB \subseteq M$. Suppose $A \not\subseteq M$. Since M is maximal, the only ideal strictly containing M is R itself, so $A + M = R$. Write $1 = a + m$ for some $a \in A$ and $m \in M$. For any $b \in B$,

$$b = 1 \cdot b = (a + m)b = ab + mb \in AB + MB \subseteq M + M = M.$$

Hence $B \subseteq M$, so M is prime. $\square$

Note: The converse is false: (0) is a prime ideal in $\mathbb{Z}$ but not a maximal ideal.

Lecture 9

Lemma 4 (6.22). *If R is a Noetherian ring, then each ideal contains a finite product of prime ideals.*

Proof. Let $\mathcal{F}$ be the family of ideals of R that do not contain any finite product of prime ideals. We want to show $\mathcal{F} = \emptyset$.

Suppose for a contradiction that $\mathcal{F} \neq \emptyset$. Since R is Noetherian, by Theorem 6.13(iii) the family $\mathcal{F}$ has a maximal element, say A .

We claim A is not a prime ideal. Indeed, any prime ideal P contains the one-factor product P itself, so no prime ideal can belong to $\mathcal{F}$.

Since A is not prime, there exist ideals B and C with $BC \subseteq A$, but $B \not\subseteq A$ and $C \not\subseteq A$. Then $B + A \supsetneq A$ and $C + A \supsetneq A$, so by maximality of A in $\mathcal{F}$, neither $B + A$ nor $C + A$ belongs to $\mathcal{F}$. Hence both contain finite products of prime ideals. But then $(B + A)(C + A)$ also contains a finite product of prime ideals, and

$$(B + A)(C + A) \subseteq BC + BA + AC + A^2 \subseteq A.$$

So A contains a finite product of prime ideals, contradicting $A \in \mathcal{F}$. Hence $\mathcal{F} = \emptyset$. $\square$

Theorem 40 (6.23 — Hilbert's Basis Theorem). *Let R be a Noetherian ring. Then the polynomial ring $R[x]$ is also Noetherian.*

Proof. Suppose for a contradiction that $I \subseteq R[x]$ is a left ideal that is not finitely generated. We recursively construct a sequence of polynomials as follows. Set

$$J_0 = (0) \subseteq R[x].$$

Since I is not finitely generated, $I \setminus J_0 \neq \emptyset$. Pick $f_0 \in I \setminus J_0$ of minimal degree (exists since degrees are non-negative integers). For $n \geq 1$, given $f_0, \dots, f_{n-1}$, let

$$J_n = (f_0, \dots, f_{n-1}) \subseteq R[x]$$

be the left ideal in $R[x]$ generated by $f_0, \dots, f_{n-1}$. Since I is not finitely generated, $I \setminus J_n \neq \emptyset$, so we may pick $f_n \in I \setminus J_n$ of minimal degree.

Since $J_n \subsetneq J_{n+1}$, the set $I \setminus J_{n+1} \subsetneq I \setminus J_n$, so $\deg(f_n)$ is non-decreasing in n .

For each n , let $a_n \in R$ denote the leading coefficient of f_n . Consider the ascending chain of left ideals in R :

$$L_1 \subseteq L_2 \subseteq \dots, \quad \text{where } L_n = (a_0, \dots, a_{n-1}) \subseteq R.$$

Since R is Noetherian, this chain stabilises: there exists $N \in \mathbb{N}$ such that $L_N = L_{N+1}$, that is,

$$a_N \in L_N = (a_0, \dots, a_{N-1}) \subseteq R.$$

So we may write $a_N = \sum_{i < N} u_i a_i$ for some $u_i \in R$. Now define

$$g = \sum_{i < N} u_i x^{\deg(f_N) - \deg(f_i)} f_i \in J_N.$$

The exponents $\deg(f_N) - \deg(f_i) \geq 0$ since degrees are non-decreasing. The leading term of $u_i x^{\deg(f_N) - \deg(f_i)} f_i$ is $u_i a_i x^{\deg(f_N)}$, so the leading term of g is

$$\left(\sum_{i < N} u_i a_i \right) x^{\deg(f_N)} = a_N x^{\deg(f_N)},$$

which equals the leading term of f_N . Hence $f_N - g$ has degree strictly less than $\deg(f_N)$.

Since $f_N \notin J_N$ and $g \in J_N$, we have $f_N - g \notin J_N$. Since $f_N, g \in I$, we have $f_N - g \in I$. Thus $f_N - g \in I \setminus J_N$.

- If $\deg(f_N) > 0$: then $f_N - g \in I \setminus J_N$ has degree strictly less than $\deg(f_N)$, contradicting the minimality of $\deg(f_N)$.

- If $\deg(f_N) = 0$: then all degrees are 0, so $f_N - g = 0$, meaning $f_N = g \in J_N$, contradicting $f_N \in I \setminus J_N$.

In both cases we reach a contradiction, so I must be finitely generated and $R[x]$ is Noetherian. $\square$

Definition 18 (7.1 — Nilpotent elements and ideals). *Let R be a ring.*

- An element $a \in R$ is nilpotent if $a^n = 0$ for some $n \in \mathbb{N}$.
- A left ideal I of R is a nil ideal if every element of I is nilpotent.
- A left ideal I of R is nilpotent if $I^n = 0$ for some $n \in \mathbb{N}$, where I^n denotes the set of all finite sums of products $i_1 i_2 \cdots i_n$ with each $i_k \in I$.

Every nilpotent ideal is nil (if $I^n = 0$ then any $a \in I$ satisfies $a^n \in I^n = 0$), but the converse fails in general.

Theorem 41 (7.5). *If J is a nil left ideal in an Artinian ring R , then J is nilpotent.*

Proof. Suppose for a contradiction that $J^k \neq 0$ for all $k \in \mathbb{N}$. The family $(J, J^2, J^3, \dots)$ is a descending chain of left ideals of R . Since R is Artinian, this family has a minimal element, say $B = J^m$ for some m . Then

$$B^2 = J^{2m} \subseteq J^m = B,$$

and by minimality of B we have $B^2 = B$.

Now consider the family

$$\mathcal{F} = \{A \mid A \text{ is a left ideal of } R, A \subseteq B, BA \neq 0\}.$$

Since $B^2 = B \neq 0$, we have $B \in \mathcal{F}$, so $\mathcal{F} \neq \emptyset$. Since R is Artinian, $\mathcal{F}$ has a minimal element A .

Since $BA \neq 0$, there exists $a \in A$ with $Ba \neq 0$. Note that $Ba = \{ba \mid b \in B\}$ is a left ideal contained in A , and

$$B(Ba) = B^2 a = Ba \neq 0,$$

so $Ba \in \mathcal{F}$. By minimality of A we conclude $Ba = A$.

Hence there exists $b \in B$ with $ba = a$. Applying b repeatedly gives $b^i a = a$ for all $i \in \mathbb{N}$. But $b \in B \subseteq J$ and J is nil, so $b^n = 0$ for some $n \in \mathbb{N}$. Then $a = b^n a = 0$, contradicting $Ba \neq 0$.

Therefore $J^k = 0$ for some $k \in \mathbb{N}$, i.e. J is nilpotent. $\square$

Lemma 5 (7.6). *Let R be a Noetherian ring. Then the sum of nilpotent left ideals in R is a nilpotent left ideal.*

Proof. Let $B = \sum_{i \in \Lambda} A_i$ be a sum of nilpotent left ideals. Since R is Noetherian, B is finitely generated as a left ideal, say $B = (x_1, \dots, x_m)$. Each x_j lies in a finite sum of the A_i 's, so B is contained in a finite sum $A_1 + \cdots + A_n$ (reindexing if necessary), and hence $B = A_1 + \cdots + A_n$.

It remains to show B is nilpotent. For each i , let $n_i \in \mathbb{N}$ be such that $A_i^{n_i} = 0$. Set $N = n_1 + \cdots + n_n$. Any product of N elements from $B = A_1 + \cdots + A_n$, when expanded, is a sum of products of the form $a_{i_1} a_{i_2} \cdots a_{i_N}$ with each $a_{i_k} \in A_{i_k}$. By pigeonhole, among the N indices $i_1, \dots, i_N \in \{1, \dots, n\}$, at least one value j appears at least n_j times. The consecutive factors from A_j multiply to give an element of $A_j^{n_j} = 0$. Hence every such product is 0, so $B^N = 0$. $\square$

Lecture 10

Theorem 42 (7.7). *Let R be a Noetherian ring having no non-zero nilpotent ideals. Then R has no non-zero nil ideals.*

(Proof non-examinable.)

Theorem 43 (7.8). *Let N be a nil ideal in a Noetherian ring R . Then N is nilpotent.*

Proof. Let T be the sum of all nilpotent ideals in R . By Lemma 7.6, T is itself nilpotent, say $T^k = 0$.

We claim R/T has no non-zero nilpotent ideals. Suppose A/T is a nilpotent ideal in R/T , say $(A/T)^m = 0$. Then $(A^m + T)/T = 0$, so $A^m \subseteq T$. Since $T^k = 0$, we get $A^{mk} \subseteq T^k = 0$, so A is nilpotent. Hence $A \subseteq T$, giving $A/T = 0$.

Since R/T is Noetherian (a quotient of a Noetherian ring) and has no non-zero nilpotent ideals, Theorem 7.7 gives that R/T has no non-zero nil ideals. Now $(N + T)/T$ is a nil ideal in R/T (it is nil since N is nil and T is nilpotent hence nil), so $(N + T)/T = 0$, which means $N \subseteq T$. Since T is nilpotent, N is nilpotent. $\square$

Lemma 6 (7.9). *Let A be a minimal left ideal in a ring R . Then either $A^2 = 0$ or $A = Re$ for some idempotent $e \in R$.*

Proof. Assume $A^2 \neq 0$.

Step 1: Find $e \in A$ with $ea = a$. Since $A^2 \neq 0$, there exists $a \in A$ with $Aa \neq 0$. The set $Aa = \{ca \mid c \in A\}$ is a left ideal contained in A (since A is a left ideal and $a \in A$). By minimality of A , we have $Aa = A$. In particular, there exists $e \in A$ with $ea = a$.

Step 2: Show $e^2 = e$. We compute $(e^2 - e)a = e(ea) - ea = ea - ea = 0$. To conclude $e^2 = e$, define the left annihilator of a in A :

$$\text{Ann}_A(a) := \{c \in A \mid ca = 0\}.$$

This is a left ideal of R contained in A . Since $ea = a \neq 0$, we have $e \notin \text{Ann}_A(a)$, so $\text{Ann}_A(a) \neq A$. By minimality of A , we conclude $\text{Ann}_A(a) = 0$. Since $e^2 - e \in A$ and $(e^2 - e)a = 0$, we get $e^2 - e \in \text{Ann}_A(a) = 0$, hence $e^2 = e$.

Step 3: Show $A = Re$. Since $e \in A$ and A is a left ideal, $Re \subseteq A$. Since $e \neq 0$ (as $ea = a \neq 0$), we have $Re \neq 0$. By minimality of A , $Re = A$. $\square$

Theorem 44 (7.10 — Wedderburn–Artin). *Let R be a left (or right) Artinian ring with unity and no non-zero nilpotent ideals. Then*

$$R \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k)$$

for some $n_1, \dots, n_k \in \mathbb{N}$ and division rings $D_1, \dots, D_k$.

Proof. We build the result from two claims, then combine them.

Claim 1. *Each non-zero left ideal A of R is of the form Re for some idempotent $e \in R$.*

By Lemma 7.9, every minimal left ideal is either nilpotent or of the form Re for some idempotent e . Since R has no non-zero nilpotent ideals, every minimal left ideal is of the form Re .

Now let A be any non-zero left ideal. Since R is Artinian, A contains a minimal left ideal, hence a non-zero idempotent. Thus the family

$$\mathcal{F} = \{R(1 - f) \cap A \mid f \text{ a non-zero idempotent in } A\}$$

is non-empty, and by the Artinian condition has a minimal member $R(1 - e) \cap A$.

Subclaim (*non-examinable*). $R(1 - e) \cap A = 0$.

Invoking the subclaim: for any $a \in A$, $a(1 - e) \in R(1 - e) \cap A = 0$, so $a = ae \in Re$. Hence $A \subseteq Re \subseteq A$, giving $A = Re$.

Claim 2. $R = Re_1 \oplus \cdots \oplus Re_n$ is a finite direct sum of minimal left ideals.

Let S be the sum of all minimal left ideals in R ; since R is Artinian it has at least one minimal left ideal, so $S \neq 0$. By Claim 1, $S = Re$ for some idempotent e .

Suppose $R(1 - e) \neq 0$. Then $R(1 - e)$ is a non-zero left ideal, so by the Artinian condition it contains a minimal left ideal A . But $A \subseteq S = Re$, giving $A \subseteq Re \cap R(1 - e)$. For any $x \in Re \cap R(1 - e)$, write $x = ae = b(1 - e)$ for some $a, b \in R$. Then $xe = ae^2 = ae = x$, and also $xe = b(1 - e)e = b(e - e^2) = 0$, so $x = 0$. Hence $Re \cap R(1 - e) = 0$, giving $A = 0$, a contradiction.

Therefore $R(1 - e) = 0$. Since R has unity, $1 - e = 0$, i.e. $e = 1$, so $S = R$.

Thus R is the sum of all its minimal left ideals. By Theorem 5.9 there exists a subfamily $(A_i)_{i \in \Lambda'}$ such that $R = \bigoplus_{i \in \Lambda'} A_i$. To see this is finite, write $1 = e_{i_1} + \cdots + e_{i_n}$ as a finite sum with $e_{i_j} \in A_{i_j}$. Then for any $r \in R$, $r = r \cdot 1 = re_{i_1} + \cdots + re_{i_n}$, so $R = Re_{i_1} \oplus \cdots \oplus Re_{i_n}$. After reindexing, $R = Re_1 \oplus \cdots \oplus Re_n$.

Conclusion. From Claim 2, $R = Re_1 \oplus \cdots \oplus Re_n$. Among $Re_1, \dots, Re_n$, select a maximal pairwise non-isomorphic subfamily; after reindexing call it $N_1 = Re_1, \dots, N_k = Re_k$. For each j , let n_j be the number of summands isomorphic to N_j , so that

$$R = \underbrace{(N_1 \oplus \cdots \oplus N_1)}_{n_1} \oplus \cdots \oplus \underbrace{(N_k \oplus \cdots \oplus N_k)}_{n_k}.$$

By Schur's Lemma 5.6, any non-zero homomorphism between simple modules is an isomorphism. Therefore $\text{Hom}_R(N_i, N_j) = 0$ whenever $i \neq j$ (since $N_i \not\cong N_j$), and $D_j := \text{Hom}_R(N_j, N_j)$ is a division ring. Invoking Theorem 4.30 on the decomposition above, the off-diagonal blocks vanish and we get

$$\text{Hom}_R(R, R) \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k).$$

It remains to identify $\text{Hom}_R(R, R)$ with R . Every $\varphi \in \text{Hom}_R(R, R)$ is determined by $b := \varphi(1)$, since $\varphi(r) = \varphi(r \cdot 1) = r\varphi(1) = rb$. So every endomorphism is right multiplication ρ_b by some $b \in R$. However composition reverses order: $(\rho_b \circ \rho_c)(1) = \rho_b(c) = cb$, which is the product $b \cdot_{\text{op}} c$ in R^{op} . Hence $\text{Hom}_R(R, R) \cong R^{\text{op}}$ as rings, giving

$$R^{\text{op}} \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k).$$

Passing to opposite rings: the opposite of a direct sum is the direct sum of the opposites, $(M_n(D))^{\text{op}} \cong M_n(D^{\text{op}})$, and D^{op} is again a division ring. Relabelling D_j^{op} as D_j gives $R \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k)$. $\square$

Theorem 45 (7.11). *Let R be a left (or right) Artinian ring with unity and no non-zero nilpotent ideals. Then R is also right and left Artinian, and right and left Noetherian.*

(Non-examinable. Follows from Wedderburn–Artin and the fact that matrix rings over division rings are Noetherian and Artinian on both sides.)

Theorem 46 (7.13 — Maschke). *If G is a finite group, then*

$$\mathbb{C}[G] \cong M_{n_1}(\mathbb{C}) \oplus \cdots \oplus M_{n_k}(\mathbb{C})$$

for some $n_1, \dots, n_k \in \mathbb{N}$.

(Non-examinable. Follows from Wedderburn–Artin: one shows $\mathbb{C}[G]$ has no non-zero nilpotent ideals and is Artinian, then uses that the only division ring containing $\mathbb{C}$ in its centre as an algebraically closed field is $\mathbb{C}$ itself.)

Theorem 47 (7.14). *Let R be a ring with unity. The following are equivalent:*

- (i) *R is left Artinian with no non-zero nilpotent ideals;*
- (ii) *R is left Artinian with no non-zero nil ideals;*
- (iii) *R is a finite direct sum of minimal left ideals;*
- (iv) *every left ideal of R is of the form Re for some idempotent e ;*
- (v) *R is a finite direct sum of matrix rings over division rings.*

(Non-examinable.)

Definition 19 (7.15 — Semisimple Artinian ring). *A ring satisfying any of the equivalent conditions of Theorem 7.14 is called a semisimple Artinian ring.*

Lecture 11 — Smith Normal Form over a PID

Throughout this chapter R is a principal ideal domain with unity.

Definition 20 (8.1 — Row module, column module). *Let $A \in M_{m \times n}(R)$. The submodule of R^n generated by the m rows of A is called the row module of A , denoted $\mathcal{R}(A)$. The submodule of R^m generated by the n columns of A is called the column module of A , denoted $\mathcal{C}(A)$.*

Since $\mathcal{R}(A)$ and $\mathcal{C}(A)$ are finitely generated submodules of the free modules R^n and R^m respectively, both are free by Lemma 5.25.

Definition 21 (8.2 — Row rank, column rank). *Let $A \in M_{m \times n}(R)$. The rank of $\mathcal{R}(A)$ is called the row rank of A , and the rank of $\mathcal{C}(A)$ is called the column rank of A .*

Theorem 48 (8.3). *Let $A \in M_{m \times n}(R)$ and let P, Q be invertible $m \times m$ and $n \times n$ matrices over R respectively. Then row rank and column rank of PAQ equal those of A .*

(Proof non-examinable.)

Definition 22 (8.4/8.6 — Elementary operations and matrices). *Let $A \in M_{m \times n}(R)$. There are three elementary row operations, each achieved by left-multiplying A by an invertible elementary matrix (column operations are achieved by right-multiplying):*

- (i) Swap rows i and j : multiply on the left by $S_{ij} = I - E_{ii} - E_{jj} + E_{ij} + E_{ji}$. Here $S_{ij}^{-1} = S_{ij}$.
- (ii) Scale row i by invertible $\alpha \in R$: multiply on the left by $M_i(\alpha) = I + (\alpha - 1)E_{ii}$. Here $M_i(\alpha)^{-1} = M_i(\alpha^{-1})$.
- (iii) Add $\alpha \in R$ times row j to row i : multiply on the left by $C_{ij}(\alpha) = I + \alpha E_{ij}$. Here $C_{ij}(\alpha)^{-1} = C_{ij}(-\alpha)$.

Since each elementary matrix is invertible, any sequence of elementary row and column operations transforms A into an equivalent matrix PAQ .

Definition 23 (8.7 — Matrix equivalence). Two matrices $A, B \in M_{m \times n}(R)$ are equivalent if there exist invertible matrices $P \in M_m(R)$ and $Q \in M_n(R)$ such that $B = PAQ$.

Theorem 49 (8.8 — Smith Normal Form). Let $A \in M_{m \times n}(R)$ where R is a PID. Then A is equivalent to a matrix of the form

$$\begin{pmatrix} a_1 & & & \\ & \ddots & & \\ & & a_r & \\ & & & 0 \end{pmatrix},$$

where $a_1, \dots, a_r \neq 0$ and $a_1 \mid a_2 \mid \dots \mid a_r$.

Proof. Define the length $\ell(a)$ of a non-zero $a \in R$ to be the number of prime factors in the factorisation $a = p_1 \cdots p_k$ counted with multiplicity, with the convention $\ell(u) = 0$ for units.

Step 1: bring a minimal-length entry to position $(1, 1)$. If $A = 0$ the result is immediate. Otherwise, among all non-zero entries pick a_{ij} with minimal $\ell(a_{ij})$. Row and column swaps bring it to position $(1, 1)$.

Step 2: clear the first row. Fix any column $k \geq 2$. If $a_{11} \mid a_{1k}$, write $a_{1k} = a_{11}q$ and apply $C_k - qC_1$, zeroing the $(1, k)$ entry without changing a_{11} . If $a_{11} \nmid a_{1k}$, let $d = \gcd(a_{11}, a_{1k})$ and write $a_{11} = ds$, $a_{1k} = dt$. By Bézout there exist $u, v \in R$ with $us + vt = 1$, so the matrix $\begin{pmatrix} u & t \\ v & -s \end{pmatrix}$ is invertible (determinant -1). Right-multiplying A by the $n \times n$ matrix with this 2×2 block in the $(1, k)$ -positions and the identity elsewhere replaces the $(1, 1)$ entry by $a_{11}u + a_{1k}v = d$ and the $(1, k)$ entry by $a_{11}t - a_{1k}s = 0$. Since d properly divides a_{11} , we have $\ell(d) < \ell(a_{11})$. Repeating across all columns clears the first row.

Step 3: clear the first column. The identical argument applied to rows (left-multiplying by the analogous block matrices) clears all entries below $(1, 1)$.

Step 4: termination. Clearing the first column may reintroduce non-zero entries in the first row, requiring another pass of Step 2. However this only occurs when a_{11} does not divide some first-column entry, which triggers the Bézout case and strictly reduces $\ell(a_{11})$. Since $\ell(a_{11}) \geq 0$ is a non-negative integer, the alternating process terminates after at most $\ell(a_{11})$ restarts, yielding

$$P_1 A Q_1 = \begin{pmatrix} a_1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & A_1 & \\ 0 & & & \end{pmatrix},$$

where A_1 is an $(m-1) \times (n-1)$ submatrix.

Step 5: induction. Apply the same procedure to A_1 and continue. By induction on $m+n$ we obtain $PAQ = \text{diag}(a_1, \dots, a_r, 0, \dots, 0)$.

Step 6: enforce $a_1 \mid a_2 \mid \dots \mid a_r$. Suppose $a_1 \nmid a_2$. Add row 2 to row 1; the first row becomes $(a_1, a_2, 0, \dots, 0)$. Applying Step 2 replaces a_1 by $\gcd(a_1, a_2)$, strictly reducing $\ell(a_1)$. Repeating forces $a_1 \mid a_2$, and similarly $a_1 \mid a_i$ for all i . Repeating the argument for a_2 in place of a_1 , and so on, gives $a_i \mid a_{i+1}$ for all $i = 1, \dots, r-1$. $\square$

Definition 24 (8.9 — Invariant factors). *The non-zero diagonal entries $a_1, \dots, a_r$ of the Smith normal form of A are called the invariant factors of A . They are unique up to unit multipliers, and two $m \times n$ matrices are equivalent if and only if they have the same invariant factors.*

Theorem 50 (8.10). *Let A be an $m \times n$ matrix over a PID. Then the row rank of A equals the column rank of A .*

Proof. Choose invertible P, Q such that PAQ is in Smith normal form (Theorem 8.8). By Theorem 8.3, multiplying by invertible matrices preserves row and column rank, so

$$\text{row rank}(A) = \text{row rank}(PAQ) = r = \text{col rank}(PAQ) = \text{col rank}(A),$$

where r is the number of non-zero diagonal entries. $\square$

Definition 25 (8.11 — Rank). *The common value of the row rank and column rank of A is called the rank of A , denoted $\text{rank}(A)$.*

Lecture 12 — Finitely Generated Modules over a PID

Theorem 51 (9.1 — Decomposition Theorem). *Let R be a principal ideal domain, and let M be a finitely generated R -module. Then*

$$M \cong R^s \oplus R/Ra_1 \oplus \dots \oplus R/Ra_r,$$

where $s, r \in \mathbb{N} \cup \{0\}$, the a_i are non-zero non-units in R , and $a_i \mid a_{i+1}$ for $i = 1, \dots, r-1$.

Proof. Since M is finitely generated, by Lemma 5.24 there is a surjective R -homomorphism $\phi_0 : R^n \rightarrow M$, so $M \cong R^n/K$ where $K = \ker \phi_0$. Since K is a submodule of the free module R^n , by Lemma 5.25 K is free of rank $m \leq n$. Let $\phi : R^m \xrightarrow{\sim} K$ be an isomorphism and $\{e_1, \dots, e_m\}$ the standard basis of R^m . Writing $\phi(e_j) = (a_{1j}, \dots, a_{nj})^\top \in R^n$, the map ϕ is left-multiplication by the matrix $A = (a_{ij}) \in M_{n \times m}(R)$, so $K = AR^m$.

By Theorem 8.8 (Smith Normal Form), there exist invertible $P \in M_n(R)$ and $Q \in M_m(R)$ such that $PAQ = \text{diag}(a_1, \dots, a_k, 0, \dots, 0)$ with $a_1 \mid \dots \mid a_k$. Since Q is invertible, $\{Qv \mid v \in R^m\} = R^m$, so $AR^m = AQR^m$. Left-multiplication by the invertible P is an automorphism of R^n sending $K = AQR^m$ to $P(AQR^m) = PAQR^m$, so $R^n/K \cong R^n/PAQR^m$. Therefore

$$M \cong \frac{R^n}{PAQR^m} \cong \frac{R}{Ra_1} \oplus \dots \oplus \frac{R}{Ra_k} \oplus R^{n-k},$$

where the last isomorphism is by Theorem 4.33, using that the image of R^m under $\text{diag}(a_1, \dots, a_k, 0, \dots, 0)$ is $Ra_1 \oplus \dots \oplus Ra_k \oplus 0 \oplus \dots \oplus 0 \subseteq R^n$. Deleting the zero summands

(those where a_i is a unit, giving $R/Ra_i = 0$), renaming the remaining non-unit diagonal entries $a_1, \dots, a_r$, and setting $s = n - k$ gives

$$M \cong \frac{R}{Ra_1} \oplus \cdots \oplus \frac{R}{Ra_r} \oplus R^s. \quad \square$$

Definition 26 (9.2 — Torsion element). *Let R be a principal ideal domain and M an R -module. An element $x \in M$ is called a torsion element if there exists a non-zero $r \in R$ such that $rx = 0$.*

Definition 27 (9.4 — Torsion-free element). *Let R be a principal ideal domain and M an R -module. A non-zero element $x \in M$ is called a torsion-free element if $rx = 0$ for $r \in R$ implies $r = 0$.*

Theorem 52 (9.6). *Let R be a principal ideal domain and M an R -module. Then*

$$\text{Tor } M := \{x \in M \mid x \text{ is a torsion element}\}$$

is a submodule of M .

Proof. We verify the three conditions of Definition 4.8.

Non-empty. For any non-zero $r \in R$, $r \cdot 0 = 0$, so $0 \in \text{Tor } M$.

Closed under subtraction. Let $x, y \in \text{Tor } M$, say $rx = 0$ and $sy = 0$ for non-zero $r, s \in R$. Then

$$rs(x - y) = s(rx) - r(sy) = 0.$$

Since R is a PID it is an integral domain, so $rs \neq 0$, hence $x - y \in \text{Tor } M$.

Closed under scalar multiplication. Let $x \in \text{Tor } M$ with $rx = 0$ for some non-zero $r \in R$, and let $a \in R$. Then $r(ax) = a(rx) = 0$, so $ax \in \text{Tor } M$. $\square$

Theorem 53 (9.7). *Let M be a finitely generated module over a principal ideal domain R . Then*

$$M = F \oplus \text{Tor } M,$$

where $F \cong R^s$ for some non-negative integer s , and $\text{Tor } M \cong R/Ra_1 \oplus \cdots \oplus R/Ra_r$ with the a_i non-zero non-units satisfying $a_1 \mid \cdots \mid a_r$.

Proof. By Theorem 9.1, $M \cong R^s \oplus R/Ra_1 \oplus \cdots \oplus R/Ra_r$. Set $F \cong R^s$ and $T \cong R/Ra_1 \oplus \cdots \oplus R/Ra_r$, so $M = F \oplus T$. We show $T = \text{Tor } M$.

$T \subseteq \text{Tor } M$. Since $a_1 \mid a_2 \mid \cdots \mid a_r$, we have $a_r \in Ra_i$ for each i , so a_r annihilates every summand R/Ra_i . Hence $a_r T = 0$, meaning every element of T is torsion, so $T \subseteq \text{Tor } M$.

$\text{Tor } M \subseteq T$. Let $x \in \text{Tor } M$ and write $x = x_1 + x_2$ with $x_1 \in F$ and $x_2 \in T$. Since $T \subseteq \text{Tor } M$, the element x_2 is torsion, and therefore so is $x_1 = x - x_2$. Say $rx_1 = 0$ for some non-zero $r \in R$. Under the isomorphism $F \cong R^s$, x_1 corresponds to some $(y_1, \dots, y_s) \in R^s$, and $r(y_1, \dots, y_s) = 0$ gives $ry_i = 0$ for each i . Since R is an integral domain and $r \neq 0$, each $y_i = 0$, so $x_1 = 0$. Thus $x = x_2 \in T$. $\square$

Theorem 54 (9.8). *Let R be a principal ideal domain. Then any non-zero ideal $P \neq R$ is prime if and only if it is maximal.*

Proof. ($\Leftarrow$) Every maximal ideal is prime by Theorem 6.21.

($\Rightarrow$) Let $P = (a)$ be a non-zero prime ideal. Suppose for a contradiction that P is not maximal, so there exists an ideal M with $P \subsetneq M \subsetneq R$. Since R is a PID, $M = (b)$

for some $b \in R$. Then $(a) \subseteq (b)$ implies $a = bx$ for some $x \in R$. Since $b \notin (a)$ and (a) is prime, the relation $a = bx \in (a)$ forces $x \in (a)$, say $x = ay$ for some $y \in R$. Then $a = bx = bay$, and since R is an integral domain and $a \neq 0$, we may cancel to get $1 = by$. But then b is a unit, so $M = (b) = R$, a contradiction. $\square$

Corollary 7. *Let R be a principal ideal domain and P a non-zero prime ideal. Then R/P is a field.*

Proof. Since P is a non-zero prime ideal in the PID R , Theorem 9.8 gives that P is maximal. It remains to show R/P is a field, i.e. that every non-zero element $a + P \in R/P$ is invertible. Since $a \notin P$, the ideal $Ra + P$ strictly contains P . As P is maximal, $Ra + P = R$, so there exist $r \in R$ and $p \in P$ with $ra + p = 1$. Then $(r + P)(a + P) = ra + P = 1 + P$, so $a + P$ is invertible. $\square$

Theorem 55 (9.9 — Uniqueness of Decomposition). *Let R be a principal ideal domain and M a finitely generated R -module. Suppose*

$$M \cong R^s \oplus R/Ra_1 \oplus \cdots \oplus R/Ra_u \quad \text{and} \quad M \cong R^t \oplus R/Rb_1 \oplus \cdots \oplus R/Rb_v,$$

where the a_i and b_j are non-zero non-units with $a_1 \mid \cdots \mid a_u$ and $b_1 \mid \cdots \mid b_v$. Then $s = t$, $u = v$, and $Ra_i = Rb_i$ for all i .

Proof. *Step 1:* $s = t$. By Theorem 9.7, both decompositions yield $M = F \oplus \text{Tor } M$ with $F \cong R^s$ and $F' \cong R^t$ respectively. Then

$$R^s \cong F \cong M/\text{Tor } M \cong F' \cong R^t,$$

so $s = t$ by Theorem 5.21. Let $T = \text{Tor } M$; it remains to show $u = v$ and $Ra_i = Rb_i$.

Step 2: $u = v$. For an R -module X and a prime $p \in R$, define $X_p := \{x \in X \mid px = 0\}$. By the corollary to Theorem 9.8, R/Rp is a field, so T_p is a vector space over R/Rp . From the first decomposition of T ,

$$\left(\frac{R}{Ra_i} \right)_p = \begin{cases} R/Rp & \text{if } p \mid a_i, \\ 0 & \text{if } p \nmid a_i, \end{cases}$$

so $\dim_{R/Rp}(T_p)$ equals the number of a_i divisible by p . Now choose a prime p dividing a_1 ; since $a_1 \mid a_i$ for all i , p divides every a_i , giving $\dim_{R/Rp}(T_p) = u$. Applying the same reasoning to the second decomposition, the dimension also equals the number of b_j divisible by p , which is at most v , so $u \leq v$. By symmetry (choosing a prime dividing b_1), $v \leq u$, hence $u = v$.

Step 3: $Ra_u = Rb_u$. Since $a_i \mid a_u$ for all i , the element a_u annihilates every summand in the first decomposition, so $a_u T = 0$. Applied to the second decomposition, $a_u \cdot (R/Rb_u) = 0$, which means $a_u \in Rb_u$, so $Ra_u \subseteq Rb_u$. By symmetry $Rb_u \subseteq Ra_u$, hence $Ra_u = Rb_u$.

The remainder of the proof is non-examinable. It remains to show $Ra_i = Rb_i$ for $i < u$ by downward induction on i . The idea is: assuming $Ra_i = Rb_i$ for $k \leq i \leq u$, one multiplies both decompositions of T by a carefully chosen element $x \in R$ built from a prime p and the exact p -adic valuation of a_u . This zeros out the first $k - 1$ summands in one decomposition but not the other, producing two decompositions of xT with a different number of non-zero summands — contradicting the fact (established by the dimension argument in Step 2) that any two such decompositions must have equally many non-zero summands. $\square$

Theorem 56 (9.10 — Fundamental Theorem of Finitely Generated Abelian Groups). *Let A be a finitely generated abelian group. Then*

$$A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z},$$

where $s \geq 0$, the a_i are non-zero non-units in $\mathbb{Z}$, and $a_1 \mid a_2 \mid \cdots \mid a_r$. This decomposition is unique.

Proof. Every abelian group is a $\mathbb{Z}$ -module (with $n \cdot a$ meaning $a + \cdots + a$ or its inverse), and $\mathbb{Z}$ is a PID. So Theorem 9.1 applies directly with $R = \mathbb{Z}$, giving a decomposition $A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z}$ where each $\mathbb{Z}/a_i\mathbb{Z} = R/Ra_i$ is a cyclic torsion summand. Uniqueness of s and of the ideals $\mathbb{Z}a_i$ (hence the a_i up to sign) then follows from Theorem 9.9.

If A is finite, then $s = 0$: the summand $\mathbb{Z}^s$ contains elements of infinite order whenever $s \geq 1$, but a finite group has no such elements. $\square$

Lecture 13 — Rational Canonical Form

Definition 28 (The $F[x]$ -module structure on V). *Let F be a field, V a finite-dimensional vector space over F , and $T \in \text{Hom}_F(V, V)$ a linear map with matrix A relative to some basis. We make V into an $F[x]$ -module by defining, for any polynomial $f(x) = a_0 + a_1x + \cdots + a_mx^m \in F[x]$ and any $v \in V$,*

$$f(x) \cdot v := a_0v + a_1(Tv) + \cdots + a_m(T^mv).$$

This extends the F -action on V so that $x \cdot v = Tv$. Since any basis $\{v_1, \dots, v_n\}$ of V over F also generates V as an $F[x]$ -module, V is a finitely generated $F[x]$ -module. Since $F[x]$ is a PID (as F is a field), the structure theorem of Lecture 12 applies.

Theorem 57 (10.1). *Let $T \in \text{Hom}_F(V, V)$, and let $f_1(x), \dots, f_n(x)$ be the invariant factors of $A - xI$, where A is a matrix of T . Then*

$$V \cong \frac{F[x]}{(f_1(x))} \oplus \cdots \oplus \frac{F[x]}{(f_n(x))}.$$

(Proof non-examinable.) The key steps are: define the surjective $F[x]$ -homomorphism $\phi : F[x]^n \rightarrow V$ by $e_i \mapsto v_i$, then show $\ker \phi = (A - xI)F[x]^n$, giving $V \cong F[x]^n / (A - xI)F[x]^n$. Since invertible changes of basis preserve this quotient up to isomorphism, one may apply Smith Normal Form (Theorem 8.8) to put $A - xI$ in diagonal form $\text{diag}(f_1(x), \dots, f_n(x))$, after which Theorem 4.33 gives the direct sum decomposition.

Remark 2 (10.2). *The polynomials $f_1(x), \dots, f_n(x)$ are called the invariant factors of A (or of T). After discarding any unit factors, the remaining non-unit invariant factors are called the invariant factors of the $F[x]$ -module V .*

Definition 29 (10.3 — T -cyclic space). *Let $T \in \text{Hom}_F(V, V)$. If V , regarded as an $F[x]$ -module via T , is cyclic, then V is called a T -cyclic space and T is called a cyclic endomorphism of V .*

Theorem 58 (10.4). *Let W be a submodule of the $F[x]$ -module V , and let $T \in \text{Hom}_F(V, V)$ satisfy $TW \subseteq W$. Then W is a T -cyclic subspace if and only if there exists $w \in W$ such that $\{w, Tw, \dots, T^{k-1}w\}$ is a basis of W for some $k \in \mathbb{N}$.*

Proof. ($\Rightarrow$) Suppose W is T -cyclic, generated by w , so $W = F[x] \cdot w$ in the sense of Definition 4.13. Since $\dim V = n$ is finite, the vectors $w, Tw, T^2w, \dots, T^nw$ must be linearly dependent over F . Let T^rw be the first of these linearly dependent on the preceding ones, so $\{w, Tw, \dots, T^{r-1}w\}$ are linearly independent and

$$T^rw = a_0w + a_1(Tw) + \dots + a_{r-1}(T^{r-1}w), \quad a_i \in F.$$

We claim every T^iw is a linear combination of $\{w, Tw, \dots, T^{r-1}w\}$. This is clear for $i \leq r-1$ and holds for $i = r$ by the equation above. For $i > r$, apply T^{i-r} to the equation to get

$$T^iw = a_0(T^{i-r}w) + \dots + a_{r-1}(T^{i-1}w),$$

and each term on the right is expressible in terms of $\{w, \dots, T^{r-1}w\}$ by the induction hypothesis. Since $W = F[x] \cdot w$, every element of W is of the form

$$f(x) \cdot w = (b_0 + b_1x + \dots + b_mx^m) \cdot w = b_0w + b_1(Tw) + \dots + b_m(T^mw),$$

using the $F[x]$ -module action from the definition. Since every T^iw lies in $\text{span}\{w, \dots, T^{r-1}w\}$, this set spans W . Combined with linear independence, $\{w, Tw, \dots, T^{r-1}w\}$ is a basis of W with $k = r$.

($\Leftarrow$) Suppose $\{w, Tw, \dots, T^{k-1}w\}$ is a basis of W . Any element $c_0w + c_1(Tw) + \dots + c_{k-1}(T^{k-1}w)$ of W equals $(c_0 + c_1x + \dots + c_{k-1}x^{k-1}) \cdot w \in F[x] \cdot w$ by the module action, so $W \subseteq F[x] \cdot w$. Since $w \in W$ and W is closed under the $F[x]$ -action, $F[x] \cdot w \subseteq W$. Hence $W = F[x] \cdot w = (w)$, so W is T -cyclic by Definition 10.3. $\square$

Definition 30 (10.5 — Companion matrix). *Let $f(x) = a_0 + a_1x + \dots + a_{k-1}x^{k-1} + x^k$ be a monic polynomial over F . The companion matrix of $f(x)$ is the $k \times k$ matrix*

$$C(f) = \begin{pmatrix} 0 & 0 & \dots & 0 & -a_0 \\ 1 & 0 & \dots & 0 & -a_1 \\ 0 & 1 & \dots & 0 & -a_2 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -a_{k-1} \end{pmatrix}.$$

Definition 31 (10.7 — Minimal polynomial of T on W). *Let W be a T -cyclic subspace generated by w . Since $F[x]$ is a PID, the annihilator $\text{Ann}(w) = \{f(x) \in F[x] \mid f(x) \cdot w = 0\}$ is a principal ideal $(f(x))$ for a unique monic generator $f(x)$. This $f(x)$ is called the minimal polynomial of T on W .*

Theorem 59 (10.6). *Let $T \in \text{Hom}_F(V, V)$ and let W be a T -cyclic subspace of V with basis $\{w, Tw, \dots, T^{k-1}w\}$. Then the matrix of T on W is the companion matrix of the minimal polynomial $f(x)$ of T on W .*

Proof. Since $\{w, Tw, \dots, T^{k-1}w\}$ is a basis of W , the vector T^kw is a linear combination of these basis elements, say

$$T^kw = a_0w + a_1(Tw) + \dots + a_{k-1}(T^{k-1}w), \quad a_i \in F.$$

In terms of the $F[x]$ -module action ($x \cdot v = Tv$), this says $(a_0 + a_1x + \dots + a_{k-1}x^{k-1} - x^k) \cdot w = 0$, so the polynomial $g(x) := a_0 + a_1x + \dots + a_{k-1}x^{k-1} - x^k$ lies in $\text{Ann}(w) = (f(x))$.

Since $F[x]$ is a PID, $\text{Ann}(w)$ is a principal ideal $(f(x))$ for some monic $f(x)$ of degree s . The inclusion $g(x) \in (f(x))$ gives $s \leq k$. On the other hand, $f(x) \cdot w = 0$ means $T^s w$

is a linear combination of $\{w, Tw, \dots, T^{s-1}w\}$. Since $\{w, Tw, \dots, T^{k-1}w\}$ are linearly independent, this forces $s \geq k$. Hence $s = k$.

Since $g(x) \in (f(x))$, write $g(x) = c \cdot f(x)$ for some $c \in F$. Both g and f are monic of degree k with leading coefficient -1 and 1 respectively, so $c = -1$ and $f(x) = -g(x) = x^k - a_{k-1}x^{k-1} - \dots - a_1x - a_0$.

The matrix of T on W relative to $\{w, Tw, \dots, T^{k-1}w\}$ has columns given by the images of each basis vector: $T(T^i w) = T^{i+1}w$ for $i < k-1$, contributing 1s on the subdiagonal, and the last column records the coefficients $a_0, \dots, a_{k-1}$ of $T^k w$. By Definition 10.5, this is exactly the companion matrix of $f(x)$. $\square$

Lecture 14

Theorem 60 (10.8 — Rational Canonical Form). *Let V be a finite-dimensional vector space over a field F and let $T \in \text{Hom}_F(V, V)$. Then there exists a basis of V , as an $F[x]$ -module, with respect to which the matrix of T is*

$$\begin{pmatrix} B_1 & & \\ & \ddots & \\ & & B_r \end{pmatrix},$$

where B_i is the companion matrix of a monic polynomial $f_i(x)$, for $i = 1, \dots, r$, such that $f_1(x) \mid f_2(x) \mid \dots \mid f_r(x)$. This matrix is called the rational canonical form of T , and it is unique.

Proof. Step 1: V is a torsion $F[x]$ -module. Since $\dim_F V = n$, we have $\dim_F \text{Hom}_F(V, V) = n^2$. So the $n^2 + 1$ elements $I, T, T^2, \dots, T^{n^2}$ must be linearly dependent over F . Let T^k be the first element linearly dependent on the preceding ones, so

$$T^k = a_0 I + a_1 T + \dots + a_{k-1} T^{k-1}, \quad a_i \in F.$$

Applying both sides to any $v \in V$ and using the $F[x]$ -module action $x \cdot v = Tv$ gives

$$(a_0 + a_1 x + \dots + a_{k-1} x^{k-1} - x^k) \cdot v = 0.$$

Since this polynomial is nonzero, every $v \in V$ is a torsion element (Definition 9.2), so $V = \text{Tor } V$.

Step 2: Decompose V using Theorem 9.1. V is a finitely generated $F[x]$ -module: any F -basis of V also generates V as an $F[x]$ -module, since $F[x]$ is a PID. Since $V = \text{Tor } V$, the free part in Theorem 9.1 vanishes, giving

$$V = V_1 \oplus \dots \oplus V_r,$$

where each $V_i \cong F[x]/(f_i(x))$ is a cyclic $F[x]$ -module and $f_1(x) \mid f_2(x) \mid \dots \mid f_r(x)$.

Step 3: Represent T on each V_i . Each V_i is cyclic as an $F[x]$ -module, hence a T -cyclic subspace (Definition 10.3). By Theorem 10.6, there exists a basis $\mathcal{B}_i$ of V_i with respect to which $T|_{V_i}$ has matrix B_i , the companion matrix of $f_i(x)$.

Step 4: Assemble. Since $V = V_1 \oplus \dots \oplus V_r$, the union $\mathcal{B} = \mathcal{B}_1 \cup \dots \cup \mathcal{B}_r$ is a basis of V , and the matrix of T with respect to $\mathcal{B}$ is $\text{diag}(B_1, \dots, B_r)$.

Uniqueness. The uniqueness of the decomposition subject to the divisibility chain, established in Theorem 9.9, implies the polynomials $f_i(x)$ — and hence the companion matrices B_i — are uniquely determined by T . $\square$

Definition 32 (11.1 — Bi-additive map). *Let M , N , and P be abelian groups. A map $f : M \times N \rightarrow P$ is called bi-additive if*

$$f(m_1 + m_2, n) = f(m_1, n) + f(m_2, n) \quad \text{and} \quad f(m, n_1 + n_2) = f(m, n_1) + f(m, n_2)$$

for all $m, m_1, m_2 \in M$ and $n, n_1, n_2 \in N$.

Definition 33 (11.3 — Balanced map). *Let M be a right R -module and N a left R -module. A map $f : M \times N \rightarrow P$ is called balanced if it is bi-additive and satisfies*

$$f(mr, n) = f(m, rn)$$

for all $m \in M$, $n \in N$, and $r \in R$.

Definition 34 (11.5 — Tensor product). *A tensor product of a right R -module M and a left R -module N is a pair (T, ϕ) , where T is an abelian group and $\phi : M \times N \rightarrow T$ is a balanced map, satisfying the following universal property: for every abelian group P and every balanced map $f : M \times N \rightarrow P$, there exists a unique group homomorphism $\bar{f} : T \rightarrow P$ such that $\bar{f} \circ \phi = f$.*

Remark 3 (11.6 — Uniqueness of the tensor product). *If (T', ϕ') is another tensor product of M and N , then there exists an isomorphism $\alpha : T \rightarrow T'$ such that $\phi' = \alpha \circ \phi$. Thus the tensor product is unique up to isomorphism. We write $M \otimes_R N$ for the tensor product and $m \otimes n$ for $\phi(m, n)$.*

Theorem 61 (11.7 — Existence of the tensor product). *Let M be a right R -module and N a left R -module. The tensor product $M \otimes_R N$ exists. Concretely, it is realised as $T(M, N) = A(M, N)/B(M, N)$, where $A(M, N)$ is the free abelian group with basis $M \times N$, and $B(M, N)$ is the subgroup generated by all elements of the form*

$$\begin{aligned} &(m_1 + m_2, n) - (m_1, n) - (m_2, n), \\ &(m, n_1 + n_2) - (m, n_1) - (m, n_2), \\ &(mr, n) - (m, rn), \end{aligned}$$

for all $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, and $r \in R$.

(Proof non-examinable.) The balanced map $\phi : M \times N \rightarrow T(M, N)$ is the restriction of the canonical quotient map $\pi : A(M, N) \rightarrow A(M, N)/B(M, N)$ to $M \times N$. Given any balanced $f : M \times N \rightarrow P$, one extends f to a homomorphism $\bar{f} : A(M, N) \rightarrow P$ using the free basis; since f is balanced, $\bar{f}$ vanishes on $B(M, N)$ and hence descends to a homomorphism $\bar{f} : T(M, N) \rightarrow P$. Uniqueness follows because $\{\phi(m, n)\}$ generates $T(M, N)$.